VIETNAM NATIONAL UNIVERSITY, HANOI
UNIVERSITY OF ENGINEERING AND TECHNOLOGY



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ATTRIBUTE-GUIDED FEATURE LEARNING FOR HIERARCHICAL FINE-GRAINED IMAGE CLASSIFICATION OF BIRDS

BACHELOR'S THESIS

Major: Information Technology

HANOI - 2026

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Supervisor: Dr. Le Duc Trong

Co-Supervisor: Dr. Do Thanh Cong

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Nguyen Huong Giang

Declaration

I am Nguyen Huong Giang, a student of class QH-2022-I/CQ-I-IT20 at VNU University of Engineering and Technology. I hereby declare that this thesis is my own work and has been carried out under the supervision of Dr. Le Duc Trong and Dr. Do Thanh Cong.

All sources of information, data, and materials used in this thesis have been properly acknowledged and cited. This work has not been submitted for any other degree or qualification at this or any other institution.

Hanoi, May 18, 2026

Student

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Abstract

Fine-grained bird image classification is challenging because visually similar species often differ only in subtle traits such as bill shape, wing pattern, head pattern, breast color, or tail shape. In a hierarchical classification setting, the model must also predict higher taxonomic levels such as family and order, while maintaining consistency across the taxonomy. This thesis studies hierarchical bird classification on CUB-200-2011 and investigates whether semantic attribute information can improve an H-CAST-based hierarchical classifier.

This thesis proposes an attribute-guided extension of H-CAST. The proposed method keeps H-CAST as the image-based hierarchical backbone and adds an attribute-guided learning module to the species-level branch. CUB attributes are organized into semantic groups, and each active group is supervised by a group-specific attribute classifier. The learned attribute-oriented features are fused with the species class token to improve fine-grained species prediction. The model uses class-level CUB attributes only as auxiliary supervision during training, while inference remains image-based. In addition, this thesis introduces Hard-Sibling Discriminative Attribute Selection, which selects attributes that help distinguish visually similar sibling species within the same family. A diversity regularization term is also studied to reduce redundancy among attribute-group attention patterns.

Experiments are conducted on the official CUB-200-2011 train/test split. Compared with the H-CAST rerun, the proposed attribute-guided H-CAST improves Full-Path Accuracy from 82.86% to 84.36%, species accuracy from 84.55% to 85.83%, family accuracy from 95.22% to 95.62%, order accuracy from 98.74% to 99.03%, and weighted Average Precision from 86.90% to 88.00%. Tree-based InConsistency Error is also reduced from 4.35% to 3.62%. Ablation studies show that attribute supervision and species-level fusion are both important, while diversity regularization improves species accuracy and wAP at a moderate weight but remains sensitive to the loss configuration. Prediction-level analysis further shows that the proposed model corrects 236 species-level errors made by the H-CAST rerun, including 194 intra-family species confusions.

Overall, the results demonstrate that semantic attribute guidance can improve H-CAST-based hierarchical bird classification, especially at the species level. However, diagnostic attention analysis shows that the model should not be interpreted as a precise attribute localization method, suggesting future work with stronger object- or part-level supervi-

sion.

Keywords: Fine-grained visual classification, hierarchical image classification, bird species recognition, visual attributes, H-CAST.

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List of Abbreviations

Abbreviation	Full Term	Description
CE	Cross-Entropy	A standard classification loss used for supervised learning.
CLS	Class Token	A learnable token used to aggregate representation for classification in transformer-based models.
CUB	Caltech-UCSD Birds	A fine-grained bird image dataset used in this thesis, specifically CUB-200-2011.
DDP	Distributed Data Parallel	A PyTorch training mechanism for distributed multi-GPU training.
FGVC	Fine-Grained Visual Classification	The task of distinguishing visually similar subordinate categories, such as bird species.
HSDAS	Hard-Sibling Discriminative Attribute Selection	The proposed attribute selection strategy that prioritizes attributes useful for distinguishing visually similar sibling species.
KL	Kullback-Leibler Divergence	A divergence measure often used to compare probability distributions.
MLP	Multi-Layer Perceptron	A feed-forward neural network module used for feature transformation or classification.
ViT	Vision Transformer	A transformer-based neural network architecture for image recognition.

Chapter 1

INTRODUCTION

1.1 Context and Motivation

Fine-grained visual classification is a specialized task in computer vision that involves identifying sub-categories within a single entry-level category. Unlike general object recognition, which differentiates between disparate classes such as automobiles and landscapes, FGVC requires the model to distinguish between highly similar categories, such as specific bird species, car models, or aircraft variants [21, 38]. Among these, bird image classification is widely regarded as one of the most challenging benchmarks due to the biological complexity and visual similarity inherent in the avian domain [44]. The primary difficulty arises from high intra-class variance, where birds of the same species appear different due to lighting, posture, or life stages, and low inter-class variance, where distinct species share nearly identical plumage patterns and morphology [21, 43].

The challenge becomes more severe when the number of candidate species is large. Given a single bird image, a non-expert observer may not be able to identify the exact species directly because many possible species look similar. However, the recognition process can become easier if the observer can describe a few meaningful visual traits, such as bill shape, wing color, head pattern, breast pattern, tail shape, or crown color. These attributes provide intermediate semantic evidence that can help narrow down candidate species and distinguish visually similar birds, especially species that belong to the same family.

Recent prompt-based and language-assisted fine-grained recognition methods reflect a similar intuition. Instead of relying only on direct class prediction, such methods use visual descriptions, attribute-related prompts, or question-answering procedures to extract discriminative semantic cues before reasoning about fine-grained categories [22]. This suggests that visual attributes can serve as useful intermediate information between raw image features and fine-grained class decisions.

At the same time, bird recognition is naturally hierarchical. Bird species are organized according to biological taxonomy, where a species belongs to a family and a family belongs

to an order. Therefore, a practical bird classification model should not only predict the fine-grained species label, but also produce predictions at higher taxonomic levels such as family and order. Hierarchical prediction is useful because coarse-level categories can organize the search space, while fine-level species prediction requires more detailed visual evidence. However, even when the higher-level family is correctly identified, distinguishing between visually similar species within the same family can remain difficult.

CUB-200-2011 provides a suitable setting for studying this problem because it contains 200 bird species and rich visual attribute annotations describing bird appearance [32]. These attributes include part-based and semantic properties such as bill shape, wing color, breast pattern, head pattern, tail shape, and crown color. Since many visually similar bird species differ in such subtle traits, attributes can provide useful semantic supervision for fine-grained hierarchical bird classification.

In this thesis, attributes are not used as manual inputs at inference time. Instead, they are used as auxiliary supervision during training. The central motivation is to investigate whether a model can learn more discriminative species-level representations by using visual attributes as semantic guidance, while still predicting species, family, and order labels from the image alone during inference.

To study this question, this thesis builds upon H-CAST, a recent visually consistent hierarchical classification framework [25]. H-CAST is selected as the backbone because it provides fine-to-coarse visual representations for hierarchical prediction. The proposed method extends this backbone with attribute-guided learning at the species level, where subtle semantic traits are most useful for distinguishing visually similar species.

1.2 Problem Statement

This thesis addresses the problem of attribute-guided fine-grained hierarchical bird image classification. Given a single bird image, the model must predict labels at multiple taxonomic levels, including order, family, and species. The task is challenging for two main reasons. First, the number of fine-grained species classes is large enough to make direct species recognition challenging. Second, many bird species, especially species from the same family, are visually similar and differ only in small local traits such as bill shape, wing pattern, head pattern, breast color, or tail shape.

Formally, for an input bird image x_i, the model is required to predict a hierarchical label

tuple:

$$(\hat{y}_i^o, \hat{y}_i^f, \hat{y}_i^s),$$

where $\hat{y}_i^o$, $\hat{y}_i^f$, and $\hat{y}_i^s$ denote the predicted order, family, and species labels, respectively. Following the CUB-200-2011 hierarchical setting, the taxonomy contains 13 orders, 38 families, and 200 species.

In addition to image labels, CUB-200-2011 provides visual attribute annotations that describe semantic bird traits. This thesis studies how such attribute information can be used during training to guide the learning of discriminative visual representations. The goal is to improve fine-grained species recognition, particularly among visually similar species within the same family, while maintaining accurate and consistent hierarchical predictions.

During inference, the model receives only the input image and does not require manually provided attribute labels. Therefore, the problem is not to classify birds using ground-truth attributes as test-time input, but to learn from attribute supervision during training so that the image-based model becomes better at hierarchical fine-grained recognition.

1.3 Research Gap and Research Questions

Fine-grained hierarchical bird classification remains challenging because the model must predict labels at multiple taxonomic levels while also distinguishing a relatively large number of visually similar species. In CUB-200-2011, many bird species belong to the same family and share highly similar global appearance. Their differences may appear only in small visual traits such as bill shape, wing pattern, head pattern, breast color, tail shape, or crown color. Therefore, image-level class supervision alone may not provide sufficient guidance for learning the subtle semantic evidence needed for species-level discrimination.

Recent prompt-based and language-assisted fine-grained recognition methods suggest that semantic visual traits can serve as useful intermediate evidence when direct category recognition is difficult [22]. Instead of predicting the fine-grained category directly, such methods often use attribute descriptions, visual questions, or semantic prompts to reason about possible classes. This motivates the question of whether visual attributes can also be useful in a supervised hierarchical bird classification setting.

CUB-200-2011 provides rich visual attribute annotations [32], but it remains unclear how these attributes should be used in a hierarchical image classification model while keeping

inference image-only. Directly using ground-truth attributes as test-time input would change the problem setting, because such attributes are not available in normal inference. Therefore, this thesis focuses on using attributes as training-time auxiliary supervision, so that the model learns more discriminative image-based representations without requiring manual attribute labels at inference time.

Another gap concerns attribute selection. Although CUB provides 312 attributes, not all of them are equally useful for distinguishing visually similar species. Some attributes may be common across many species, redundant with other attributes, or weakly related to the most difficult species-level confusions. This is especially important for species within the same family, where useful attributes should help distinguish hard sibling species rather than only describe general bird appearance.

Finally, it is necessary to identify which components actually contribute to the benefit of attribute guidance. Attribute-guided learning may improve performance because of auxiliary semantic supervision, because of species-level feature fusion, because of cross-attention-based attribute feature extraction, or simply because of additional model capacity. Therefore, the thesis evaluates these components through controlled ablation studies, including direct-fusion alternatives and variants without auxiliary attribute supervision.

The internal Sibling-IVAS baseline in this thesis refers to a node-vs-siblings attribute selection method based on positive occurrence rate, negative occurrence rate, and information value.

To address these gaps, this thesis studies the following research questions:

- **RQ1: Overall effectiveness of attribute guidance.** Can visual attributes improve fine-grained hierarchical bird classification when they are used as auxiliary supervision during training, while inference remains image-only?
- **RQ2: Role of attribute-guided fusion and semantic supervision.** Does the improvement come from species-level feature fusion, auxiliary attribute prediction supervision, or the combination of both?
- **RQ3: Role of cross-attention.** Does attribute-query cross-attention provide additional benefit compared with a simpler direct-fusion design that does not use cross-attention?
- **RQ4: Attribute selection strategy.** Does Hard-Sibling Discriminative At-

tribute Selection (HSDAS) provide a more effective attribute subset than using all attributes, equally sized random subsets, or the internal Sibling-IVAS statistical baseline?

- **RQ5: Role of diversity regularization.** How does attribute attention diversity regularization affect species accuracy, Full-Path Accuracy, weighted Average Precision, and taxonomy consistency?

- **RQ6: Error behavior.** In which types of prediction errors does the attribute-guided model improve over the image-only hierarchical baseline, and do the improvements mainly occur in intra-family species confusions?

1.4 Objectives

The primary objective of this thesis is to develop and evaluate an attribute-guided method for supervised fine-grained hierarchical bird image classification on CUB-200-2011. Specifically, the study aims to:

- Use CUB visual attributes as training-time auxiliary semantic supervision while preserving image-only inference.

- Organize the 312 CUB visual attributes into semantic groups and formulate attribute prediction as group-wise classification.

- Design an attribute-guided species-level feature learning module that uses attribute-oriented representations to support fine-grained species prediction.

- Instantiate the proposed attribute-guided approach on top of H-CAST, a visually consistent hierarchical classification backbone.

- Compare attribute-query cross-attention with a simpler direct-fusion baseline that does not use cross-attention.

- Study attribute usage and selection strategies, including FullAttr-312, Sibling-IVAS-188, HSDAS-174, HSDAS-199, and Random-199.

- Analyze the effects of attribute supervision, feature fusion, cross-attention, diversity regularization, and attribute selection on species accuracy, family accuracy, order

accuracy, Full-Path Accuracy, weighted Average Precision, TICE, and attribute prediction accuracy.

- Conduct prediction-level and diagnostic analyses to understand which types of errors are corrected and to examine how the proposed module changes model behavior.

1.5 Scope of the Study

The scope of this research is limited to the CUB-200-2011 dataset. The study focuses on supervised fine-grained hierarchical bird image classification and does not address open-set recognition, zero-shot recognition, or prompt-based inference. Although recent prompt-based and language-assisted methods motivate the usefulness of semantic attributes for fine-grained reasoning, this thesis does not use language prompts, large language models, or manually provided attribute descriptions at inference time.

The CUB visual attributes are used only during training as auxiliary supervision. In the main setting of this thesis, class-level attribute annotations are used to supervise the group-wise attribute prediction heads. Therefore, all images from the same species share the same class-level attribute target. This setting is useful for injecting species-level semantic priors, but it does not model image-specific attribute visibility caused by pose, occlusion, viewpoint, or background variation.

This thesis does not aim to perform attribute localization; attention and relevance maps are used only for diagnostic visualization and are not evaluated as localization results.

The proposed attribute-guided module is applied primarily to the species-level prediction branch, because species recognition requires the most fine-grained visual evidence. Family and order predictions follow the higher-level hierarchical stages of the visual backbone. The main experiments follow the official CUB-200-2011 train/test protocol used in the H-CAST benchmark. Therefore, the conclusions are limited to this dataset and should not be assumed to generalize to other hierarchical classification tasks without further evaluation.

1.6 Overview of the Proposed Approach

This thesis proposes an attribute-guided approach for fine-grained hierarchical bird image classification on CUB-200-2011. The method is implemented on top of H-CAST, a visually

consistent hierarchical classification framework. H-CAST is selected as the backbone because it provides fine-to-coarse visual representations through superpixel-based graph grouping, which is suitable for predicting different taxonomy levels such as species, family, and order.

The proposed model preserves the original H-CAST visual parsing pipeline, where image regions are represented by superpixel-based tokens and progressively grouped through graph pooling. This produces hierarchical visual representations at different granularities. The species-level representation is expected to preserve more detailed visual evidence, while higher-level representations are used for family and order prediction.

On top of this image-based hierarchical backbone, the proposed method introduces an attribute-guided module at the species-level branch. This design is motivated by the fact that many species-level confusions occur between visually similar birds within the same family, where subtle traits such as bill shape, wing pattern, head pattern, breast color, or tail shape may be important. Instead of using attributes as additional inputs at inference time, the model uses CUB attributes only as auxiliary supervision during training.

The proposed module organizes selected CUB attributes into semantic groups, such as bill shape, wing color, breast pattern, head pattern, tail shape, and crown color. Each active attribute group is represented by a learnable query. These attribute-group queries attend to the graph-node features from the species-level stage and produce attribute-oriented features. A group-specific classifier predicts the retained attribute category within each active group, providing auxiliary semantic supervision.

The resulting attribute-oriented features are aggregated using an attention-pooling mechanism conditioned on the species class token. The aggregated attribute representation is concatenated with the species-level visual representation and passed through a fusion module before the final species classifier. Family and order predictions are still produced from the original higher-level backbone stages, while the attribute-guided branch primarily aims to improve fine-grained species discrimination.

To evaluate whether attribute-query cross-attention is necessary, this thesis also studies a direct-fusion baseline that removes the cross-attention module. In this baseline, group-wise attribute features are generated directly from the species class token and fused with the species representation. This comparison helps determine whether the proposed cross-attention mechanism provides additional benefit beyond simple feature fusion and auxiliary attribute supervision.

Since not all CUB attributes are equally useful for distinguishing visually similar bird species, this thesis also investigates attribute selection. In particular, Hard-Sibling Discriminative Attribute Selection (HSDAS) selects attributes that help distinguish a species from its sibling species within the same family, with emphasis on hard siblings that have similar attribute profiles. HSDAS is compared with using all attributes, an internal node-vs-siblings statistical selection baseline called Sibling-IVAS, and random attribute subsets.

For the cross-attention-based model, this thesis also evaluates an attribute attention diversity regularization term. This term is intended to reduce redundancy among macro-part-level attribute attention patterns. Because attribute localization is not an objective of this thesis, attention visualizations are used only as diagnostic evidence for model behavior rather than as localization results.

1.7 Contributions

The main contributions of this thesis are as follows:

- **Attribute-guided fine-grained hierarchical bird classification.** This thesis studies how CUB visual attributes can be used as training-time auxiliary supervision to improve fine-grained hierarchical bird image classification while preserving image-only inference.
- **Group-wise semantic attribute supervision.** Instead of treating all 312 CUB attributes as a single flat vector, this thesis organizes attributes into semantic groups and formulates attribute prediction as group-wise classification.
- **Attribute-guided species-level fusion.** The proposed model extracts attribute-oriented features from species-level visual representations and fuses the aggregated attribute representation with the species class token before final species classification.
- **Cross-attention and direct-fusion comparison.** This thesis evaluates whether attribute-query cross-attention provides additional benefit over a simpler direct-fusion design without cross-attention.
- **Hard-Sibling Discriminative Attribute Selection.** This thesis studies HSDAS, an attribute selection strategy that prioritizes attributes useful for distinguishing visually similar sibling species within the same family.

- **Component-wise evaluation and diagnostic analysis.** The proposed method is evaluated through main comparison with an image-only hierarchical baseline, fusion and attribute-loss ablations, cross-attention versus direct-fusion comparison, attribute mask comparison, diversity-loss analysis, random attribute controls, prediction-level error analysis, and diagnostic attention/relevance analysis.

1.8 Thesis Organization

The remainder of this thesis is organized as follows:

- **Chapter 2** reviews the theoretical background and related work in fine-grained visual classification, attribute-based recognition, prompt-based fine-grained reasoning, hierarchical image classification, and H-CAST.
- **Chapter 3** presents the proposed attribute-guided hierarchical bird classification framework, including the H-CAST backbone, group-wise attribute representation, HSDAS attribute selection, attribute-guided learning module, training objective, and inference procedure.
- **Chapter 4** describes the experimental setup, evaluation metrics, main results, component ablations, direct-fusion comparison, attribute selection studies, diversity-loss analysis, prediction-level error analysis, and diagnostic attention/relevance visualizations.
- **Chapter 5** summarizes the thesis, discusses limitations, and suggests future research directions.

Chapter 2

RELATED WORK

2.1 Fine-Grained Visual Classification

Fine-grained visual classification (FGVC) refers to the task of distinguishing subordinate categories that belong to the same general visual category, such as bird species, aircraft variants, or car models [35]. Compared with generic object recognition, FGVC is more challenging because the visual differences between classes are often subtle and localized, while images from the same class may still vary significantly due to pose, viewpoint, illumination, background, and object deformation [21, 35, 43]. In bird species recognition, two species may share nearly identical global shapes and color distributions, differing only in minute visual traits such as bill shape, wing pattern, crown color, or breast pattern [11, 32, 38]. Consequently, effective FGVC systems must be capable of learning representations that simultaneously capture global object structure and localized discriminative details [38].

Early FGVC studies often relied on part annotations, bounding boxes, or explicitly designed part localization mechanisms to compare corresponding regions across images. For example, part-based R-CNN methods learn object and part detectors and use geometric constraints to construct pose-normalized representations for fine-grained categorization [41]. More recent deep learning methods reduce the dependency on manual part supervision by learning discriminative regions or feature interactions directly from image-level labels. Representative approaches include bilinear feature modeling [20], recurrent attention [9], and region proposal-based attention mechanisms [37]. These methods demonstrate that discovering local discriminative evidence is central to fine-grained recognition. However, many conventional FGVC methods are formulated as single-level fine-grained classification problems, where the output is only the finest category and higher-level taxonomic consistency is not explicitly modeled [25, 35].

This thesis focuses on fine-grained bird recognition in a hierarchical setting. Instead of predicting only the species label, the model predicts labels at multiple taxonomic levels, namely order, family, and species. This formulation is particularly relevant for bird

classification, where the taxonomy provides useful semantic structure and where mistakes between closely related species are common. The following subsections review the CUB-200-2011 dataset, discriminative region learning in FGVC, and diversity regularization methods that motivate the attribute-guided learning strategy used in this thesis.

2.1.1 Bird Species Recognition and the CUB-200-2011 Dataset

The Caltech-UCSD Birds-200-2011 dataset [32], commonly known as CUB-200-2011, is one of the most widely used benchmarks for fine-grained bird image classification. It contains 200 bird species and 11,788 images, with annotations including bounding boxes, part locations, and 312 binary visual attributes. These attributes describe semantic visual properties of birds, such as color, shape, and pattern, and are organized around bird parts such as the bill, wing, breast, belly, tail, head, and leg.

CUB-200-2011 is particularly suitable for studying attribute-guided fine-grained classification for several reasons. First, the species are visually similar and often differ only in localized traits. Second, the dataset provides rich attribute annotations that can be used as auxiliary semantic supervision. Third, the bird taxonomy can be organized into a hierarchy of order, family, and species, making it appropriate for hierarchical image classification. In this thesis, the CUB-200-2011 hierarchy follows the H-CAST setting with 13 orders, 38 families, and 200 species [25].

The attributes in CUB are useful because they provide human-interpretable descriptions of subtle visual cues. For instance, two bird species may have a similar body shape and overall color, but differ in bill shape, wing color, crown pattern, or belly pattern. These attributes can guide a model toward fine-grained traits that may be difficult to learn from class labels alone, consistent with prior attribute-guided and attribute-aware attention methods for fine-grained recognition [10, 23]. In this thesis, CUB attributes are not used as manual inputs during inference. Instead, they are used during training as auxiliary supervision for learning attribute-oriented representations. This design preserves the standard image-only inference setting while exploiting additional semantic information available in the training data.

Nevertheless, using all available attributes may not always be optimal. Some attributes may be redundant, weakly discriminative, or irrelevant for distinguishing a particular group of sibling species. This motivates the study of attribute selection strategies, including the proposed Hard-Sibling Discriminative Attribute Selection (HSDAS), which aims

to identify attributes that are useful for distinguishing taxonomically related and visually similar bird classes.

2.1.2 Discriminative Region Learning in FGVC

A central challenge in FGVC is learning where to look. Since fine-grained categories are often separated by small local differences, many methods aim to identify discriminative object regions or part-level patterns. Bilinear CNN models, for example, model pairwise interactions between local convolutional features and have been shown to be effective for texture-rich and fine-grained recognition tasks [20]. Although bilinear pooling does not explicitly require part annotations, it can capture local feature interactions that are useful for distinguishing visually similar categories.

Another line of work uses attention mechanisms to locate informative regions. RA-CNN recursively learns region attention and region-based feature representations at multiple scales, allowing the model to progressively focus on discriminative image regions [9]. Similarly, NTS-Net introduces a Navigator-Teacher-Scrutinizer framework, where informative regions are proposed and then scrutinized for final fine-grained prediction [37]. These methods highlight the importance of localized evidence for fine-grained recognition.

Despite their effectiveness, many discriminative region learning methods are designed for flat species-level classification. They aim to improve the prediction of the finest label, but they do not explicitly model taxonomy-level consistency. In contrast, hierarchical classification requires the model to produce predictions at multiple semantic levels. For example, a species prediction should be compatible with its corresponding family and order. This thesis therefore builds on the visually consistent hierarchical classification framework of H-CAST, which uses segment-based visual grouping to align fine-to-coarse predictions [25].

The proposed approach differs from conventional discriminative region learning in two ways. First, it uses hierarchical visual representations from H-CAST rather than a flat image classifier. Second, it incorporates bird attributes as auxiliary supervision, encouraging the model to learn representations related to semantic visual traits. Instead of directly predicting attributes as the final task, the model uses attribute-oriented features to support species-level classification while preserving hierarchical predictions at the family and order levels.

2.1.3 Loss-based Diversity Regularization

In addition to architecture-based region learning, another direction in FGVC encourages discriminative and diverse feature learning through loss functions. Mutual-Channel Loss (MC-Loss) is a representative example of this direction [4]. MC-Loss was proposed to learn localized discriminative features without requiring explicit part or bounding-box annotations. It consists of two main components: a discriminality component and a diversity component. The discriminality component encourages feature channels associated with a class to be individually discriminative, while the diversity component encourages those channels to focus on different spatial regions.

The intuition behind the diversity component is that a fine-grained object cannot usually be recognized from a single local cue. For bird recognition, different parts such as the head, wing, breast, bill, and tail may all provide complementary evidence. If all feature channels attend to the same most salient region, the model may ignore other useful visual traits. By encouraging feature channels to be spatially diverse, MC-Loss helps the network discover multiple discriminative regions and reduce redundant feature responses [4].

This idea is relevant to attribute-guided bird classification. In the proposed model, different attribute groups may correspond to different semantic parts or visual properties. For example, bill shape, crown color, wing pattern, and tail shape should ideally provide complementary information rather than collapsing onto the same image region. Therefore, this thesis adopts the diversity principle from MC-Loss and adapts it to the attribute-guided H-CAST setting.

It is important to note that the diversity regularization used in this thesis is not identical to the original MC-Loss. MC-Loss applies diversity regularization to class-specific convolutional feature channels. In contrast, this thesis applies a diversity-inspired regularization term to attribute-group attention patterns. The goal is to encourage different attribute groups or macro-parts to attend to complementary visual regions. Thus, the proposed loss should be understood as an adaptation of the MC-Loss diversity idea, rather than a direct reuse of the original MC-Loss formulation.

By combining attribute supervision with diversity regularization, the model is encouraged to learn attribute-oriented representations that are both discriminative and complementary. This is especially important for CUB-200-2011, where species-level differences often depend on multiple subtle visual attributes. The effect of this diversity regularization

is evaluated in later experimental chapters through ablation studies with different loss weights.

2.2 Attribute-Based and Attribute-Guided Recognition

Visual attributes provide a semantic description of object appearance. Instead of representing an object only by its category label, attributes describe human-interpretable properties such as color, shape, texture, pattern, or part configuration. In image recognition, this representation is useful because many visual categories share common properties. For example, different bird species may share attributes such as a pointed bill or striped wings, while still differing in other subtle traits. Therefore, attributes can serve as an intermediate semantic layer between low-level visual features and high-level class labels.

Attribute-based recognition has been studied as a way to make visual models more interpretable and transferable. Farhadi et al. introduced the idea of describing objects by attributes, shifting recognition from only naming object categories to also describing their visual properties [8]. Lampert et al. further showed that attributes can support recognition of unseen categories by transferring knowledge through semantic attribute descriptions [18, 19]. Although these works are often discussed in the context of zero-shot learning, the underlying idea is also relevant to supervised fine-grained recognition: attributes can guide a model toward visual traits that are shared across classes and meaningful to humans.

In this thesis, attributes are not used for zero-shot prediction. Instead, they are used as auxiliary supervision for learning better image representations. The goal is to encourage the model to encode fine-grained visual traits that are useful for distinguishing bird species. This differs from attribute-based zero-shot methods that rely on class-level attribute prototypes or semantic descriptions as part of the inference mechanism. In the proposed framework, no manually provided image-level attribute labels are required at inference time.

2.2.1 Visual Attributes as Semantic Supervision

In conventional supervised image classification, the training signal is usually provided only by class labels. This supervision tells the model which category an image belongs to, but it does not explicitly describe why the image belongs to that category. For fine-grained

classification, this limitation is important because two classes may be separated by very small differences. A class label alone may not explicitly encourage the model to focus on the bird's bill, crown, wing, tail, or breast pattern. Visual attributes provide additional semantic supervision by describing these discriminative properties.

Attributes can be used in several different ways. One common strategy is to predict attributes as intermediate semantic outputs and then use the predicted attributes for classification. The Direct Attribute Prediction and Indirect Attribute Prediction frameworks of Lampert et al. are representative examples in zero-shot recognition [18, 19]. Another strategy is to use attributes as class-level semantic embeddings. Akata et al. embed class labels in an attribute-vector space and learn image-to-label compatibility for attribute-based classification [1, 2]. Semantic Autoencoder later followed the semantic embedding direction by learning a projection from visual features to semantic space and adding a reconstruction constraint from semantic space back to visual features [16]. These methods demonstrate that attributes can define a structured semantic space for visual recognition.

Attributes can also be used as auxiliary supervision or as guidance for attention. Liu et al. proposed an attribute-guided attention localization method for fine-grained recognition, where part attribute descriptions guide the learning of local region localizers on CUB-200-2011 [23]. Their method trains localizers for parts such as head, wing, breast, and tail using part attribute descriptions, and the attribute descriptions are used during training rather than as test-time inputs. Han et al. proposed the Attribute-Aware Attention Model (A^3M), which learns local attribute representations and global category representations simultaneously through attribute-guided and category-guided attention modules [10]. Huynh and Elhamifar proposed dense attribute-based attention for fine-grained generalized zero-shot learning, where each attribute is associated with relevant image regions to obtain attribute-based features [12]. These works provide concrete examples of how attributes can guide the model toward local discriminative regions.

The auxiliary-supervision setting is particularly relevant to this thesis. Instead of using attributes as the final prediction target or requiring human-provided attributes at inference time, the proposed model uses attribute labels during training to shape the learned representation. This design is related to previous attribute-guided attention methods, but differs in two aspects. First, it is built on an H-CAST-based hierarchical visual backbone rather than a flat image classifier. Second, it uses attribute-group queries and species-level

attribute fusion to support hierarchical bird classification.

The main advantage of attribute supervision is that it introduces structured semantic information into the learning process. Instead of treating each species as an unrelated index, the model can receive additional signals about visual properties that may be shared across species. For example, attributes related to wing color, bill shape, or belly pattern may help the model learn features that are useful for multiple bird classes. This can regularize the learned representation and make it more aligned with human-interpretable visual traits.

However, attribute supervision also introduces challenges. First, not all attributes are equally useful for a specific classification problem. Some attributes may be common across many classes and therefore weakly discriminative. Second, different attributes may be redundant, especially when they describe correlated visual properties. Third, noisy or irrelevant attributes may distract the model from the main classification objective. This concern is consistent with recent work on attribute-based visual recognition showing that very large automatically generated attribute sets can contain substantial noise, and that concise attribute subsets may preserve performance while improving interpretability [36]. These challenges motivate the attribute selection component studied later in this thesis.

2.2.2 Attributes in Fine-Grained Bird Recognition

CUB-200-2011 is especially suitable for studying attributes in fine-grained bird recognition because it provides both species labels and rich visual attribute annotations [32]. The dataset contains 200 bird species and 312 binary attributes describing local and global bird appearance. These attributes include properties related to bill shape, wing color, upperparts color, underparts color, breast pattern, tail shape, head pattern, eye color, leg color, and other bird parts. Such annotations make CUB a natural benchmark for attribute-guided fine-grained classification. In this thesis, these annotations are used through the class-level continuous attribute file, where each value represents the percentage of annotators who considered an attribute present for a given species.

Bird species recognition benefits from attributes because many species are distinguished by localized traits. For instance, two species may have similar body shapes and overall color distributions, but differ in crown color, wing pattern, bill length, or tail shape. These differences are often subtle and may occupy only a small part of the image. Attribute supervision can therefore provide a useful learning signal by indicating which kinds of

visual traits are relevant for discrimination.

Prior work has also shown the importance of semantic descriptions for fine-grained visual recognition. Reed et al. studied fine-grained visual descriptions and learned joint representations between images and textual descriptions on datasets including CUB [27]. Their work demonstrates that semantic side information can be useful for representing fine-grained categories. In contrast to description-based or zero-shot recognition settings, this thesis uses the structured attribute annotations already provided by CUB as auxiliary supervision for a supervised hierarchical classification task.

The use of CUB attributes in this thesis differs from direct attribute-based classification. The model does not first predict a complete attribute vector and then classify the bird solely from that vector. Instead, attribute-related supervision is integrated into an H-CAST-based visual backbone. Attribute-group queries are used to extract attribute-oriented features from species-level graph-node representations, and these features are fused with the species-level class token. The attribute predictions act as auxiliary outputs during training, while the final inference remains image-based.

Another important issue is attribute subset selection. Since CUB contains 312 attributes, using all attributes may introduce redundancy or weakly useful supervision. Some attributes may help distinguish only a few species, while others may be common across many sibling classes. This thesis therefore compares multiple attribute usage strategies, including using all attributes, an internal node-vs-siblings statistical selection baseline, random attribute subsets, and the proposed Hard-Sibling Discriminative Attribute Selection (HSDAS). The internal statistical baseline, designed in this thesis and referred to as Sibling-IVAS-188, uses train-only image-level binary attributes and ranks attributes for each taxonomy node primarily by positive occurrence-rate difference, with Information Value used as a secondary criterion. The goal is to study whether a compact and taxonomically meaningful attribute subset can better support hierarchical fine-grained bird classification.

2.3 Hierarchical Image Classification

Hierarchical image classification [5, 13, 29] extends standard image classification by predicting labels according to a predefined class hierarchy. Instead of assigning each image to only one flat class, a hierarchical classifier can produce predictions at multiple semantic

levels, such as order, family, and species in bird recognition. This setting is useful when the label space naturally follows a taxonomy and when different levels of granularity are meaningful for different users or application scenarios.

In the context of this thesis, hierarchical classification is particularly relevant because bird categories are naturally organized taxonomically. A model that predicts only the species label may achieve fine-grained recognition, but it does not explicitly evaluate whether the predicted species is compatible with the corresponding family and order. Therefore, hierarchical classification provides a more structured formulation for bird recognition, where both fine-grained accuracy and taxonomy-level consistency are important.

2.3.1 Flat Classification vs. Hierarchical Classification

Flat classification treats all target categories as independent classes at a single semantic level [29]. In fine-grained bird classification, this usually means predicting one of the 200 species in CUB-200-2011. This formulation is simple and widely used, but it ignores the hierarchical relationships between classes. For example, if a model predicts a species from one family while its family-level prediction indicates a different family, the prediction path becomes inconsistent with the taxonomy.

Hierarchical classification, by contrast, explicitly models labels at multiple levels of abstraction. In a bird taxonomy, the model may predict the order, family, and species of an image. This formulation allows the model to represent both coarse and fine semantic information. It also makes evaluation more informative, since hierarchy-aware evaluation treats an error between two species in the same family as less severe than an error between species from different orders [3].

Existing hierarchical classification methods can be broadly divided into single-level and full-taxonomy prediction approaches. In a bottom-up single-level approach, the model predicts the finest label, and higher-level labels are inferred from the taxonomy. This guarantees a valid taxonomy path, but it relies heavily on the correctness of the fine-grained prediction. In a top-down approach, the model first predicts a coarse label and then refines the prediction toward finer labels. This can be useful when coarse-level recognition is easier, but early mistakes may propagate downward [25].

Full-taxonomy prediction aims to predict all hierarchy levels simultaneously. This is the setting used in this thesis. It provides explicit supervision at each level and allows the

model to evaluate performance at order, family, and species levels separately. However, because different heads may predict different levels independently, full-taxonomy prediction must address the problem of taxonomy consistency [25].

2.3.2 Taxonomy Consistency and Multi-Granularity Prediction

A central challenge in hierarchical classification is maintaining consistency among predictions at different levels. A prediction is taxonomically consistent if the predicted labels form a valid path in the hierarchy. For example, a predicted species should belong to the predicted family, and that family should belong to the predicted order. If the species and family predictions do not match the known taxonomy, the model produces an inconsistent prediction path.

This issue is especially important in multi-granularity fine-grained recognition. Different levels of the hierarchy may require different visual evidence. Coarse-level labels often depend on broad object-level cues, while fine-grained species labels may depend on subtle local details. H-CAST emphasizes this problem by showing that classifiers at different hierarchy levels may attend to different image regions, leading to inconsistent predictions even when semantic supervision is used [25].

Several metrics are used to evaluate hierarchical prediction. Level-wise accuracy measures the classification accuracy at each individual level, such as order accuracy, family accuracy, and species accuracy. However, level-wise accuracy alone does not fully capture whether the predictions form a valid taxonomy path. Tree-based InConsistency Error rate (TICE) measures the proportion of samples whose predicted path is inconsistent with the hierarchy. Full-Path Accuracy (FPA), on the other hand, counts a prediction as correct only when all hierarchy levels are predicted correctly. Therefore, FPA evaluates both correctness and hierarchical agreement, while TICE focuses on the validity of the predicted path [25].

Multi-granularity prediction is useful because it allows a model to provide predictions at different levels of specificity. In practical bird recognition, an image may not always contain enough detail for confident species-level prediction, but a coarse-level family or order prediction may still be meaningful. At the same time, predicting multiple levels introduces a trade-off: the model must learn features that are useful for coarse categories while preserving subtle visual cues needed for fine-grained species discrimination.

2.3.3 Representative Hierarchical Classification Methods

A number of methods have been proposed to exploit hierarchy information in image classification. Bertinetto et al. studied how class hierarchies can be used to make "better mistakes" by encouraging errors that are closer to the ground-truth class in the hierarchy [3]. Their work highlights that not all classification errors are equally severe and motivates the use of hierarchy-aware evaluation and training objectives.

For fine-grained visual classification, Chang et al. proposed FGN, which reformulates traditional FGVC as a multi-granularity recognition problem [5]. Instead of predicting only the finest category, FGN considers a coarse-to-fine label hierarchy and uses level-specific classification heads to reduce negative transfer across granularity levels. The method also allows fine-grained features to contribute to coarse-level prediction, reflecting the observation that fine-level visual details can help recognize coarser labels.

Chen et al. proposed the Hierarchical Residual Network (HRN) for hierarchical multi-granularity classification [6]. HRN models hierarchical feature interaction by adding granularity-specific features from parent levels to child-level features through residual connections. It also uses a combinatorial loss based on the tree hierarchy to exploit label relationships and transfer hierarchical knowledge during training [6]. This approach focuses on transferring hierarchical knowledge between levels and improving feature interaction across granularities.

With the rise of Vision Transformers, Wang et al. introduced TransHP, a hierarchical prompting method for image classification [34]. TransHP learns prompt tokens representing coarse ancestor classes and injects the predicted coarse-class prompt into intermediate transformer features. The motivation is that ancestor-level information can guide the model to focus on subtle differences among descendant classes. Unlike residual CNN-based approaches, TransHP uses the transformer token mechanism to condition feature extraction on hierarchy information.

Another related direction focuses directly on prediction consistency. Wang et al. proposed Consistency-Aware Feature Learning (CAFL) for hierarchical fine-grained visual classification [33]. CAFL encodes prediction consistency constraints through forward deduction and backward induction over the label hierarchy, aiming to improve both classification accuracy and consistency. This type of method shows that enforcing consistency is important, but it mainly focuses on semantic relations between labels and prediction scores.

More recently, H-CAST introduced a visually consistent approach to hierarchical image classification [25]. Unlike methods that rely mainly on label relations or semantic losses, H-CAST builds on CAST and uses superpixel-based hierarchical visual grouping. It applies fine-to-coarse supervision at different stages of the model so that finer visual segments support fine-grained labels and coarser segments support higher-level labels. In addition, H-CAST uses Tree-Path KL Divergence to encourage semantic consistency across hierarchy levels.

H-CAST is the most directly related baseline for this thesis. It provides a strong image-only framework for hierarchical bird classification and addresses both visual and semantic consistency. However, in its CUB setting, H-CAST uses the order-family-species taxonomy but does not incorporate the CUB visual attribute annotations as explicit auxiliary supervision. This thesis builds on the H-CAST framework by introducing attribute-guided learning at the species-level branch. The goal is to investigate whether semantic attribute supervision and hard-sibling-aware attribute selection can further improve fine-grained species discrimination while preserving hierarchical prediction.

2.4 Vision Transformers and Segment-Based Visual Grouping

Vision Transformers have become an important backbone family for image recognition. Unlike convolutional neural networks, which process images through local convolutional filters and spatial downsampling, Vision Transformers represent an image as a sequence of tokens and use self-attention to model relationships among them [7]. This token-based formulation provides a flexible framework for visual representation learning, but standard ViT models usually rely on fixed square patch tokens. Such tokens are convenient for computation, but because they are defined by a regular grid, they may cut across object boundaries and are not explicitly designed to correspond to semantic parts [14, 25].

For fine-grained and hierarchical recognition, this limitation is important. Distinguishing bird species often requires localized evidence from object parts, such as the bill, wing, head, breast, or tail. At the same time, higher-level categories such as family or order may depend on broader visual structures. Therefore, a model that can organize visual information from small parts to larger regions is naturally suitable for hierarchical bird classification. This motivates the use of segment-based visual grouping methods in CAST [14] and H-CAST [25].

2.4.1 Vision Transformers

The Vision Transformer (ViT) adapts the Transformer architecture [31] to image classification by dividing an image into fixed-size patches and treating each patch as a token [7]. A learnable class token is appended to the patch tokens, and the sequence is processed by multiple Transformer encoder blocks. Through self-attention, each token can interact with other tokens, allowing the model to capture long-range dependencies across the image.

ViT provides a simple and powerful image classification framework, but its patch-based representation has limitations for visual parsing. Since patches are defined by a regular grid, they may cut across object boundaries or merge foreground and background pixels. This can be problematic in fine-grained recognition, where the relevant visual evidence may correspond to irregularly shaped parts rather than square image regions.

Another limitation is that standard ViT maintains patch tokens as the main visual units throughout the network. Although self-attention can aggregate information globally, the model does not explicitly produce a part-to-whole segmentation hierarchy. For hierarchical classification, this means that the model may not directly align fine-grained visual cues with fine labels and broader visual regions with coarse labels. These issues motivate later segment-based extensions of ViT, including CAST and H-CAST.

2.4.2 CAST

The model is named CAST because it Concurrently learns segmentation and recognition using Adaptive Segment Tokens, modifying the standard ViT formulation by replacing fixed square patches with adaptive segment tokens based on superpixels [14]. CAST enhances the standard ViT formulation with adaptive segment tokens and graph pooling, allowing segmentation and recognition to be learned concurrently within a single model [14]. Its central idea is that segmentation should not be treated only as an independent end task, but as an internal visual parsing process that evolves with and supports image recognition. CAST learns hierarchical segmentation together with recognition using image-level recognition objectives, without requiring pixel-level segmentation labels.

CAST starts by extracting superpixels from the input image using SEEDS [30]. These superpixels serve as the finest visual units and are used to construct adaptive segment tokens. Compared with fixed patch tokens, superpixels can better follow low-level color

and boundary structure than fixed square patches, providing adaptive visual units before higher-level graph pooling. Each segment token is then processed by Transformer blocks, and graph pooling is used to progressively merge finer segments into coarser segments.

The graph pooling module constructs a fine-to-coarse hierarchy of segment tokens. At each level, finer segment tokens are softly assigned to coarser segment tokens based on feature similarity. This produces a hierarchy of segmentations and features, allowing the model to gradually move from local parts to larger object regions. The CAST paper emphasizes that this hierarchical segmentation is not trained with external segmentation labels; instead, it emerges from the recognition objective itself [14].

This design distinguishes CAST from standard ViT in two major ways. First, the visual tokens are adaptive segments rather than fixed patches. Second, the number and granularity of tokens change through graph pooling, producing a part-to-whole visual hierarchy. As a result, CAST can provide both image-level recognition and internal hierarchical segmentation. The CAST paper reports that this segment-based representation improves unsupervised part-whole discovery, semantic segmentation, image classification, and efficiency compared with ViT baselines [14].

CAST is relevant to this thesis because it provides the visual backbone on which H-CAST is built. Its fine-to-coarse segment hierarchy makes it possible to connect visual granularity with semantic granularity. In bird classification, fine segments can preserve local details such as the head, bill, or wing, while coarser segments can represent larger bird regions. This property is useful for hierarchical classification, where different taxonomy levels may require different levels of visual detail.

2.4.3 H-CAST

H-CAST extends CAST to hierarchical image classification [25]. While CAST learns hierarchical segmentation for a flat recognition objective, H-CAST introduces hierarchical supervision so that different stages of the visual hierarchy are associated with different taxonomy levels. The model is motivated by the observation that fine-grained and coarse-grained classification may rely on different but related visual cues. For example, distinguishing between visually similar bird species may require local details such as the bill, wings, or head coloration, whereas recognizing a higher-level category may require broader object-level evidence.

The key idea of H-CAST is internal visual consistency. H-CAST argues that predictions at different hierarchy levels should be grounded in coherent and related visual regions. H-CAST addresses this issue by using the fine-to-coarse visual grouping structure inherited from CAST. Fine-level visual details are progressively grouped into coarser regions, so predictions at different hierarchy levels are encouraged to rely on coherent and related image regions.

Formally, H-CAST considers an image associated with labels at multiple hierarchy levels. It uses the segment and feature representations at different levels of the CAST hierarchy, adds a classification head at each level, and applies cross-entropy supervision across all hierarchy levels. In contrast to CAST, which uses a flat supervision signal on the final class token, H-CAST applies hierarchical supervision at multiple stages of the visual parsing process [25]. This allows fine and coarse labels to guide different levels of visual representation.

In addition to visual consistency, H-CAST introduces Tree-Path KL Divergence loss to improve semantic consistency. This loss concatenates the labels from all hierarchy levels into a single distribution and aligns the concatenated outputs of the classification heads with this distribution using KL divergence. The purpose is to reduce prediction paths that violate the taxonomy, such as predicting a species whose parent family or order is inconsistent with the predicted higher-level labels [25].

The H-CAST paper evaluates hierarchical classification using both accuracy and consistency metrics. It defines Full-Path Accuracy (FPA) as the proportion of samples for which all hierarchy levels are correctly predicted, and uses Tree-based InConsistency Error rate (TICE) to measure whether predicted paths are valid in the taxonomy [25]. These metrics are also relevant to this thesis because the proposed model is evaluated not only by species accuracy, but also by family accuracy, order accuracy, FPA, and TICE.

H-CAST is the main baseline and architectural foundation of this thesis. The proposed method keeps the H-CAST visual parsing pipeline and extends it with attribute-guided learning for CUB-200-2011. Specifically, instead of changing the hierarchical segment grouping mechanism, this thesis adds an attribute-guided module at the species-level branch. The motivation is that H-CAST provides visually consistent hierarchical representations, while CUB attributes provide additional semantic supervision about fine-grained bird traits. Combining these two sources of information may help the model better distinguish visually similar sibling species.

2.5 Attribute Selection for Fine-Grained Hierarchical Classification

Attribute annotations provide useful semantic supervision for fine-grained recognition, but using all available attributes is not always optimal, since feature-selection studies emphasize the importance of removing irrelevant or redundant variables [26], and recent work on attribute-based visual recognition also suggests that concise attribute subsets can preserve performance while improving interpretability [36]. In datasets such as CUB-200-2011, the attribute space is rich and detailed, covering color, shape, pattern, and body-part-related properties [32]. However, not every attribute is equally useful for distinguishing every bird species. Some attributes may be shared by many species, some may be redundant with other attributes, and some may be weakly related to the most difficult class distinctions. Therefore, when attributes are used as auxiliary supervision, attribute selection becomes an important design choice.

In general machine learning, feature selection aims to identify a subset of informative variables while removing irrelevant or redundant ones. Classical approaches often evaluate features according to their relevance to target labels and, in some cases, their redundancy with other features. For example, the minimum Redundancy Maximum Relevance (mRMR) criterion selects features that are highly relevant to the target while being minimally redundant with already selected features [26]. Relief-based methods estimate the quality of attributes by comparing nearby samples from the same and different classes, making them sensitive to local class distinctions [15, 17].In applied scorecard and risk modeling, Weight of Evidence and Information Value are commonly used during variable binning and screening to assess the usefulness of predictive variables [28].

Although these feature selection methods provide useful general principles, fine-grained hierarchical classification introduces additional considerations. First, the most useful attributes may not be globally discriminative across all classes. An attribute that separates visually unrelated birds may be less useful for distinguishing two sibling species within the same family. Second, the hierarchical structure of the labels means that discrimination can occur at multiple levels, such as order, family, and species. Third, in fine-grained bird classification, difficult errors often occur between visually and taxonomically related classes. Therefore, an effective attribute subset should support local discrimination among related classes, rather than only global separation across the entire dataset.

This idea is related to prior work on taxonomy-aware attribute querying. Mall et al. proposed a field-guide-inspired zero-shot learning framework in which a learner asks for the most useful attributes to describe a new class [24]. Their method is motivated by real field guides, where a species can be described by a small number of informative attributes rather than requiring annotation of the complete attribute vector. In particular, their *sibling-variance* strategy selects attributes that vary within a branch of the taxonomy, showing that taxonomy-aware attribute selection can be more informative than measuring attribute variation globally. This work is closely related to the motivation of this thesis, because both approaches emphasize attributes that distinguish related classes.

More broadly, attribute selection in this thesis is also related to hierarchical feature selection. Zhao et al. proposed hierarchical feature selection with recursive regularization, where different feature subsets are selected for different nodes in a class hierarchy by using parent-child and sibling relationships [42]. This line of work shows that hierarchical class structure can be useful for selecting features, because related nodes may share common features while sibling nodes may require discriminative features that separate them. However, these methods are general feature selection frameworks and are not specifically designed for CUB bird attributes or for attribute-guided visual representation learning.

This thesis considers attribute selection in the context of attribute-guided H-CAST. Since the proposed model uses attribute groups as auxiliary supervision and species-level feature guidance, the selected attributes influence which semantic cues are emphasized during training. If too many weak or redundant attributes are included, the attribute branch may learn less useful supervision and reduce the effectiveness of attribute-guided fusion. Conversely, if too few attributes are selected, the model may lose important cues needed to distinguish similar bird species. Attribute selection is therefore studied as a trade-off between compactness, discriminative power, and semantic coverage.

Several attribute usage strategies are considered in this thesis. The simplest strategy, referred to as **FullAttr-312**, uses all 312 CUB attributes. This setting preserves the full semantic information available in the dataset, but it may also include redundant or weakly discriminative attributes.

This thesis also includes an internal statistical attribute-selection baseline, referred to as **Sibling-IVAS-188**. This baseline was implemented as an exploratory comparison method rather than adopted from a published work. It uses train-only image-level binary attribute annotations and compares each taxonomy node with a corresponding negative

set. For species-level nodes, the negative set consists of sibling species within the same family; for family-level nodes, it consists of sibling families within the same order; and for order-level nodes, it consists of the remaining orders. Attributes are ranked primarily by positive occurrence-rate difference between the target node and the negative set, with Information Value used as an additional ranking criterion. Therefore, Sibling-IVAS-188 should be understood as a taxonomy-aware statistical baseline, not as a global attribute-selection method.

To provide a stronger control, this thesis also includes **Random-199**, a random attribute subset with the same number of attributes as the main selected subset. This baseline is important because it helps determine whether any performance change is caused by the attribute selection criterion itself or merely by reducing the number of attributes. If a selected subset outperforms a random subset of the same size, this provides evidence that the selection strategy captures meaningful discriminative information.

The main attribute selection strategy studied in this thesis is **Hard-Sibling Discriminative Attribute Selection (HSDAS)**. HSDAS follows the high-level intuition of sibling-aware attribute selection: attributes should be chosen according to local taxonomic distinctions, not only according to global discriminative strength. However, the setting is different from field-guide-inspired zero-shot learning. Mall et al. use sibling-aware attribute querying to reduce the cost of expert annotation for novel classes [24]. In contrast, this thesis assumes that class-level CUB attributes are already available during training and uses HSDAS to construct an attribute supervision mask for an H-CAST-based hierarchical classifier.

Conceptually, HSDAS reflects the nature of fine-grained bird classification. Two species from the same family may share global shape and overall color but differ in localized properties such as crown color, bill shape, wing pattern, or tail shape. Attributes that distinguish these hard sibling classes are more likely to support species-level recognition than attributes that separate visually unrelated birds. Therefore, HSDAS prioritizes attributes that distinguish a taxonomic node from its sibling nodes, especially hard siblings with similar attribute profiles.

In this thesis, HSDAS is not presented as a general-purpose feature selection algorithm for all datasets. Rather, it is designed for the specific setting where a taxonomy and class-level attribute annotations are available. Its role is to select a compact set of semantic attributes that can guide an H-CAST-based classifier toward discriminative bird traits. The selected

attributes are then used to define active attribute groups for auxiliary attribute prediction and attribute-guided species-level fusion.

The study of attribute selection also supports the interpretability of the proposed approach. Since selected attributes correspond to human-readable bird traits, the resulting subset can be analyzed in terms of body parts and semantic groups. This makes it possible to inspect whether the model is guided by meaningful attributes such as bill shape, wing color, tail pattern, or breast markings. However, attribute selection does not imply that the model receives manually provided attribute labels at inference time. The attributes are used only during training as auxiliary supervision, while inference remains image-based.

Overall, attribute selection is a key component of the proposed attribute-guided hierarchical classification framework. It connects the semantic structure of CUB attributes with the taxonomic structure of bird labels. By comparing FullAttr-312, Sibling-IVAS-188, Random-199, and HSDAS-199, this thesis evaluates whether a compact and hard-sibling-aware attribute subset can better support fine-grained species recognition and hierarchical prediction than using all attributes, random subsets, or earlier internal attribute-selection baselines.

2.6 Research Gap

The previous sections show that fine-grained bird classification, attribute-based recognition, and hierarchical image classification have each been studied from different perspectives. Fine-grained visual classification methods focus on learning discriminative visual evidence for subtle category differences. Attribute-based methods use semantic visual properties to describe object appearance and provide additional supervision. Hierarchical classification methods exploit taxonomy structures to produce predictions at multiple semantic levels. However, the integration of these directions remains limited in the context of visually consistent hierarchical bird classification.

First, recent hierarchical classification methods have improved taxonomy-aware prediction by modeling relationships between coarse and fine labels. Methods such as FGN, HRN, TransHP, and H-CAST show that hierarchical information or supervision can improve multi-granularity recognition [5, 6, 25, 34]. Among them, H-CAST is particularly relevant to this thesis because it addresses both visual consistency and semantic consistency. It

builds on CAST and uses fine-to-coarse segment grouping so that classifiers at different taxonomy levels are encouraged to rely on coherent image regions [14, 25]. Nevertheless, in its CUB setting, H-CAST uses the order-family-species taxonomy but does not explicitly incorporate the rich CUB visual attribute annotations as auxiliary supervision.

Second, CUB-200-2011 provides detailed visual attributes describing bird appearance, including properties related to bill shape, wing color, breast pattern, tail shape, crown color, and other part-based traits [32]. These attributes are closely related to the subtle differences that distinguish visually similar bird species. Prior attribute-based recognition studies demonstrate that attributes can serve as semantic descriptions for visual categories and support transferable recognition [8, 19]. Other attribute-guided methods further show that attributes can guide attention or representation learning for localized visual traits [10, 12, 23]. However, these methods are not specifically designed to work with a visually consistent hierarchical segmentation model such as H-CAST. As a result, there remains a gap between attribute-guided representation learning and segment-based hierarchical visual parsing.

Third, even when attributes are available, it is unclear whether using all attributes is the best strategy for supervised hierarchical bird classification. CUB contains 312 attributes, but not all of them are equally useful for every classification decision. Some attributes may be redundant, broadly shared across many species, or weakly discriminative for distinguishing sibling species. General feature selection methods provide criteria for selecting informative variables [15, 17, 26], but they do not directly address the use of human-interpretable bird attributes inside a hierarchical visual classification model.

More closely related to this thesis, sibling-aware attribute querying has been studied in field-guide-inspired zero-shot learning. Mall et al. propose sibling-aware attribute querying strategies that prioritize attributes useful for distinguishing a class from related classes in a taxonomy, showing that local taxonomic distinctions can be more informative than global attribute variation [24]. However, their setting focuses on reducing expert annotation cost for zero-shot class descriptions. In contrast, this thesis studies a supervised hierarchical classification setting where class-level CUB attributes are already available during training. The gap addressed here is how to use hard-sibling-aware attribute selection to construct a compact attribute supervision mask for an H-CAST-based hierarchical classifier.

Fourth, when multiple attribute groups are introduced into a visual model, another chal-

lenge is how to make their learned representations complementary. If different attribute groups attend to the same visual region, the model may fail to exploit the diverse semantic information provided by attributes. Diversity-based regularization methods in fine-grained classification suggest that encouraging different features to focus on different discriminative regions can reduce redundancy [4]. In this thesis, this idea is adapted from channel-level feature diversity to attribute-group attention diversity in an H-CAST-based model.

Motivated by these gaps, this thesis investigates attribute-guided feature learning for hierarchical bird image classification. The proposed approach extends the image-only H-CAST framework by adding attribute-group supervision and species-level attribute-guided fusion. It also studies Hard-Sibling Discriminative Attribute Selection (HSDAS), which adapts the intuition of sibling-aware attribute selection to supervised H-CAST-based bird classification. Rather than claiming sibling-aware attribute selection as a completely new idea, this thesis focuses on integrating hard-sibling-aware attribute selection with attribute-guided representation learning in a visually consistent hierarchical classifier. Through comparisons with FullAttr-312, Sibling-IVAS-188, and Random-199, this thesis evaluates whether a compact, hard-sibling-aware attribute subset can better support fine-grained species recognition and hierarchical consistency on CUB-200-2011 than using all attributes, equally sized random subsets, or the internal Sibling-IVAS statistical baseline.

Chapter 3

ATTRIBUTE-GUIDED H-CAST FOR HIERARCHICAL BIRD CLASSIFICATION

3.1 Overview of the Proposed Method

This thesis proposes an attribute-guided extension of H-CAST for hierarchical bird image classification on CUB-200-2011. The original H-CAST framework performs hierarchical image classification by combining segment-based visual grouping with multi-level supervision [25]. It processes an input image through a fine-to-coarse visual hierarchy and predicts labels at different taxonomic levels, such as species, family, and order. However, in its CUB setting, H-CAST uses the order-family-species taxonomy but does not explicitly incorporate the rich visual attribute annotations provided by CUB-200-2011 as auxiliary supervision. The proposed method addresses this limitation by integrating attribute-based supervision and attribute-guided species-level feature fusion into the H-CAST pipeline.

Figure 3.1 illustrates the overall architecture of the proposed method. Given an input bird image, the H-CAST backbone first extracts hierarchical visual representations using superpixel-based segment tokens and graph pooling, following CAST and H-CAST [14, 25]. These representations are used to produce predictions at different taxonomy levels. In the original H-CAST design, the species, family, and order classifiers are attached to different stages of the visual hierarchy. In the proposed model, this hierarchical structure is preserved. The family and order predictions are still produced from the corresponding higher-level H-CAST stages, while the species-level branch is enhanced with attribute-guided learning.

The main addition of this thesis is an attribute-guided module attached to the species-level branch. This design is motivated by the nature of fine-grained bird classification. Species-level recognition often depends on subtle and localized visual traits, such as bill shape, crown color, wing pattern, breast markings, or tail shape. These traits are directly described by CUB attributes [32]. Therefore, instead of relying only on species labels, the proposed model uses attribute annotations as auxiliary supervision to encourage the

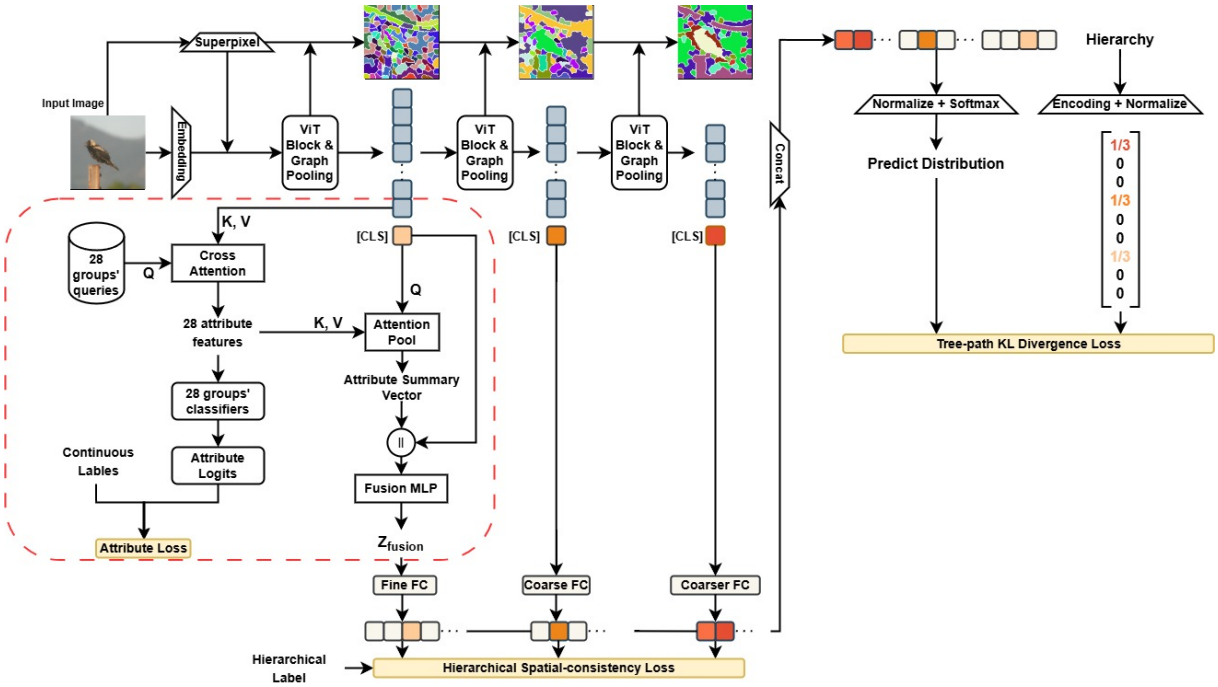


Figure 3.1: Overview of the proposed attribute-guided H-CAST framework. The H-CAST backbone extracts hierarchical visual representations from the input image. Attribute-group queries attend to species-level graph-node features and produce attribute-oriented representations. These representations are supervised by group-specific attribute prediction heads and fused with the species-level class token for final species prediction. Family and order predictions are produced from the original H-CAST hierarchy. Attribute labels are used only during training.

learned representation to encode fine-grained bird characteristics.

The attribute-guided module operates at the level of attribute groups rather than individual binary attributes. In this thesis, the 312 CUB attributes are organized into semantic groups based on their attribute names, such as bill shape, wing color, head pattern, breast color, and tail shape. For each active attribute group, the model maintains a learnable attribute-group query. These queries attend to the graph-node features from the species-level stage of H-CAST and extract attribute-oriented visual representations. Each attribute-oriented representation is then passed to a group-specific classifier, which predicts the retained attribute category within that group. These attribute predictions are used as auxiliary outputs during training.

In addition to auxiliary attribute prediction, the proposed model uses the learned attribute-oriented features to improve species classification. The attribute features are aggregated through an attention-pooling mechanism conditioned on the species-level visual represen-

tation. The resulting attribute summary is concatenated with the species class token and passed through a fusion module before the final species classifier. In this way, the species prediction is guided not only by the image representation learned by H-CAST, but also by attribute-oriented features that capture discriminative bird traits.

An important design choice is that the attribute-guided fusion is applied primarily to the species-level branch. This is consistent with the role of attributes in fine-grained recognition: attributes are expected to be most useful for distinguishing visually similar species, while family and order labels correspond to coarser taxonomic levels. Therefore, the proposed model preserves the original H-CAST predictions for family and order, and introduces attribute-guided fusion only for species prediction. Although family and order classifiers are not directly fused with attribute features, they are still trained jointly with the species branch in the hierarchical classification framework.

Since CUB provides 312 attributes, using all attributes may introduce redundancy or weakly useful supervision. To address this issue, this thesis studies several attribute selection strategies. The proposed strategy, Hard-Sibling Discriminative Attribute Selection (HSDAS), selects attributes that help distinguish a taxonomic node from its sibling nodes, especially hard siblings with similar attribute profiles. The selected attributes define the active attribute groups used by the attribute prediction heads and the attribute-guided fusion module. This allows the model to focus on a compact subset of attributes that are more relevant to fine-grained hierarchical discrimination.

The model is trained with a combination of level-wise hierarchical classification loss, Tree-Path KL divergence for semantic consistency, attribute prediction loss, and an optional attribute attention diversity loss. The hierarchical classification loss follows the H-CAST setting and supervises predictions at species, family, and order levels. The attribute prediction loss supervises the group-specific attribute heads using the selected attribute labels. The diversity loss is inspired by the diversity component of Mutual-Channel Loss [4], but is adapted to attribute-group attention maps. Its purpose is to encourage different attribute groups to attend to complementary visual regions instead of collapsing onto the same image evidence.

During inference, the model remains image-based. No manually provided attribute labels are required. Given an input image, the H-CAST backbone extracts hierarchical visual features, the attribute-group queries produce internal attribute-oriented representations, and the fused species representation is used for species prediction. The model also outputs

family and order predictions from the original H-CAST hierarchy. Thus, attributes are used to guide representation learning during training, while the final prediction pipeline remains compatible with standard image-only classification.

Overall, the proposed method combines three components: a visually consistent H-CAST backbone, attribute-guided species-level feature learning, and hard-sibling-aware attribute selection. The goal is to improve fine-grained species discrimination while preserving hierarchical prediction across species, family, and order. This design directly addresses the research gap identified in Chapter 2: H-CAST provides a strong image-only hierarchical classifier, while CUB attributes provide semantic cues that can guide the model toward discriminative bird traits.

3.2 Problem Formulation

This thesis addresses supervised hierarchical bird image classification on the CUB-200-2011 dataset. Each input sample is a bird image associated with a species label, and each species belongs to a fixed taxonomic hierarchy. In this work, the hierarchy contains three levels: order, family, and species. The species level is the finest level, while family and order represent progressively coarser semantic categories.

Let the training set be denoted as

$$\mathcal{D} = \{(x_i, y_i^s, y_i^f, y_i^o)\}_{i=1}^N, \tag{3.1}$$

where x_i is the input image, y_i^s is the species label, y_i^f is the family label, and y_i^o is the order label. The label spaces are denoted by $\mathcal{Y}^s$, $\mathcal{Y}^f$, and $\mathcal{Y}^o$, respectively. During model computation, each image is also converted into an initial superpixel partition, which is generated automatically by using SEEDS [30] and is not treated as an additional ground-truth label. For CUB-200-2011, $\mathcal{Y}^s$ contains 200 species [32]. Following the hierarchical setting used in H-CAST, these species are organized into 38 families and 13 orders [25].

The taxonomy defines parent-child mappings between levels. Let

$$\pi_f : \mathcal{Y}^s \to \mathcal{Y}^f \tag{3.2}$$

be the mapping from a species to its parent family, and let

$$\pi_o : \mathcal{Y}^f \to \mathcal{Y}^o \tag{3.3}$$

be the mapping from a family to its parent order. Therefore, for each training sample, the hierarchy satisfies

$$y_i^f = \pi_f(y_i^s), \qquad y_i^o = \pi_o(y_i^f). \tag{3.4}$$

The goal of the model is to learn a function f_θ that predicts all three taxonomy levels from an input image:

$$f_\theta(x_i) = (\hat{y}_i^s, \hat{y}_i^f, \hat{y}_i^o), \tag{3.5}$$

where $\hat{y}_i^s$, $\hat{y}_i^f$, and $\hat{y}_i^o$ are the predicted species, family, and order labels. A prediction is taxonomically consistent if the predicted labels form a valid path in the hierarchy:

$$\hat{y}_i^f = \pi_f(\hat{y}_i^s), \qquad \hat{y}_i^o = \pi_o(\hat{y}_i^f). \tag{3.6}$$

Thus, the model is expected not only to predict accurate labels at each level, but also to produce predictions that are consistent with the bird taxonomy.

In addition to taxonomy labels, CUB-200-2011 provides visual attribute annotations [32]. Let the full CUB attribute vector for a species c be denoted as

$$\mathbf{a}_c \in \mathbb{R}^{312}. \tag{3.7}$$

These attributes describe semantic bird traits such as bill shape, wing color, breast pattern, tail shape, crown color, and other part-based properties. In this thesis, attributes are used as class-level auxiliary supervision. For an image x_i with species label y_i^s, the corresponding attribute target is obtained from the class-level attribute vector:

$$\mathbf{a}_i = \mathbf{a}_{y_i^s}. \tag{3.8}$$

The 312 CUB attributes are organized into semantic groups. Let

$$\mathcal{G} = \{g_1, g_2, \ldots, g_K\} \tag{3.9}$$

denote the set of attribute groups, where each group corresponds to a bird trait type such as bill shape, wing color, head pattern, or tail shape. Each group g contains a set of attribute indices I_g. Instead of treating all 312 attributes as independent binary labels, the proposed model formulates attribute prediction as a group-wise classification problem. For each group g, the target attribute category is determined from the attribute values within that group:

$$t_{i,g} = \arg\max_{j \in I_g} a_{i,j}. \tag{3.10}$$

When an attribute selection mask is used, only a subset of attributes is retained. Let $M_g \subseteq I_g$ denote the selected attribute indices for group g. If $t_{i,g} \in M_g$, the target is remapped to a local class index:

$$\tilde{t}_{i,g} = \phi_g(t_{i,g}), \tag{3.11}$$

where $\phi_g(\cdot)$ maps an original CUB attribute index to its local index within the selected group M_g. If $t_{i,g} \notin M_g$, then $\tilde{t}_{i,g}$ is assigned an ignore label and excluded from the attribute loss. This formulation allows the model to train with compact selected attribute subsets such as HSDAS-199, while preserving the group-wise semantic structure of CUB attributes.

The proposed model therefore produces two types of outputs. The first type is the hierarchical classification output:

$$(\mathbf{z}_i^s, \mathbf{z}_i^f, \mathbf{z}_i^o), \tag{3.12}$$

where $\mathbf{z}_i^s$, $\mathbf{z}_i^f$, and $\mathbf{z}_i^o$ are the logits for species, family, and order prediction. The second type is the auxiliary attribute output:

$$\{\mathbf{z}_{i,g}^a\}_{g \in \mathcal{G}_{active}}, \tag{3.13}$$

where $\mathcal{G}_{active}$ is the set of attribute groups retained by the selected attribute mask, and $\mathbf{z}_{i,g}^a$ is the logit vector for attribute group g.

The main classification task is supervised by the species, family, and order labels. The auxiliary attribute task is supervised by the selected group-wise attribute targets. Importantly, attribute labels are used only during training. During inference, the model receives only an input image and predicts the taxonomic labels. No manually provided attribute labels are required at test time.

Overall, the problem studied in this thesis can be summarized as follows: given an input bird image and a known taxonomy, learn an image-based hierarchical classifier that predicts species, family, and order labels, while using CUB attribute annotations during training to guide the species-level visual representation. The proposed method aims to improve fine-grained species discrimination and maintain hierarchical prediction quality by combining H-CAST visual representations with attribute-guided learning.

3.3 H-CAST Backbone

The proposed model in this thesis is built on top of H-CAST, a visually consistent hierarchical image classification framework. H-CAST itself is a supervised hierarchical extension of CAST, which is a Vision Transformer-based architecture that integrates hierarchical image segmentation into the recognition process. This section describes the H-CAST backbone used in this thesis before introducing the proposed attribute-guided extension in the following sections.

3.3.1 From ViT to CAST

The standard Vision Transformer represents an image as a sequence of fixed square patch tokens [7]. Although this representation is effective for image classification, patch tokens are not necessarily aligned with object boundaries or semantic parts. This limitation is important for fine-grained bird classification, where the discriminative evidence may lie in localized and irregularly shaped regions such as the bill, wing, head, breast, or tail.

CAST addresses this issue by replacing fixed patch tokens with adaptive segment tokens [14]. Instead of dividing the image into a regular grid, CAST first extracts superpixels from the input image using SEEDS [14, 30]. These superpixels are designed to follow low-level color and boundary structure more closely than fixed square patches. In the CAST framework, segmentation is not treated as a separate task requiring pixel-level labels. Rather, hierarchical segmentation is integrated into the recognition process and learned from image-level recognition objectives [14].

Let S_0 denote the initial superpixel partition of the input image. Each superpixel corresponds to an adaptive segment token. CAST first obtains pixel-level features using convolutional layers, then averages the pixel features inside each superpixel to form the initial segment feature. Positional information is also aggregated within each superpixel. The resulting segment tokens are appended with a class token, forming the initial token representation:

$$Z_0 = [X_{class}; X_s + E_{pos}], \tag{3.14}$$

where X_s denotes the visual features of the superpixels, E_{pos} denotes the corresponding positional encodings, and X_{class} is the class token.

The key operation in CAST is graph pooling. Given segment features at a finer level, graph pooling softly assigns fine segment tokens to a smaller set of coarser segment tokens.

This operation progressively constructs a fine-to-coarse visual parsing hierarchy. Let P_l denote the soft assignment matrix from level $l-1$ to level l. The segmentation at the coarser level is obtained by propagating the previous segmentation through the assignment matrix:

$$S_l = S_{l-1} P_l. \tag{3.15}$$

In this way, CAST produces a hierarchy of segmentations and features. Finer levels preserve local details, while coarser levels aggregate larger regions and capture more global visual context.

Compared with a standard ViT, CAST differs in two important aspects. First, its visual tokens correspond to superpixel-based image segments rather than square patches. Second, the number of tokens decreases through graph pooling, allowing the model to form a part-to-whole visual hierarchy. This hierarchy is learned as an internal process for recognition, without requiring external segmentation annotations.

3.3.2 H-CAST for Hierarchical Classification

H-CAST extends CAST from flat image recognition to hierarchical image classification [25]. In a hierarchical task, an input image is associated with labels at multiple taxonomy levels. Let x denote an image and let

$$(y_1, y_2, \ldots, y_L) \tag{3.16}$$

denote its hierarchical labels, where y_1 is the coarsest label and y_L is the finest label. For CUB-200-2011, the hierarchy contains three levels: order, family, and species. Therefore, the model must predict not only the species label, but also the corresponding family and order labels.

The main motivation of H-CAST is that different taxonomy levels require different levels of visual detail. Fine-grained species recognition may require local details such as bill shape, wing markings, or head coloration. Coarser recognition, such as family or order classification, may rely on broader visual structures. However, these levels should still be visually consistent: coarse-level predictions should be grounded in regions that are related to the fine-level evidence, rather than attending to unrelated background or objects.

To support this behavior, H-CAST attaches classification heads to different levels of the CAST visual hierarchy. Let Z_l and S_l denote the feature representation and segmentation

at level l. The feature representation Z_l contains a class token Z_l^{class} and segment tokens Z_l^{seg}. For each taxonomy level l, H-CAST uses a classification head f_l, implemented as a linear classifier, on the corresponding class token:

$$\hat{\mathbf{z}}_l = f_l(Z_l^{class}). \tag{3.17}$$

Here, $\hat{\mathbf{z}}_l$ denotes the logits for the label space at level l.

H-CAST applies hierarchical supervision across all taxonomy levels. The hierarchical visual consistency loss is defined as the sum of cross-entropy losses over all levels:

$$\mathcal{L}_{HV} = \sum_{l=1}^{L} \mathcal{L}_{CE}\left(f_l(Z_l^{class}), y_l\right). \tag{3.18}$$

This differs from the original CAST setting, where supervision is applied only to the final class token for a flat recognition objective. By applying classification losses at multiple stages, H-CAST aligns different levels of the visual parsing hierarchy with different levels of the semantic taxonomy.

In the CUB setting used in this thesis, the finest prediction corresponds to species, while the coarser predictions correspond to family and order. The H-CAST backbone therefore provides three hierarchical classification outputs:

$$(\hat{\mathbf{z}}^s, \hat{\mathbf{z}}^f, \hat{\mathbf{z}}^o), \tag{3.19}$$

where $\hat{\mathbf{z}}^s$, $\hat{\mathbf{z}}^f$, and $\hat{\mathbf{z}}^o$ are the logits for species, family, and order, respectively.

3.3.3 Semantic Consistency with Tree-Path KL Divergence

Although hierarchical supervision encourages each level to make a correct prediction, independently trained classification heads may still produce inconsistent paths. For example, a model may predict a species from one family while predicting a different family at the family level. This violates the known taxonomy.

To reduce such semantic inconsistency, H-CAST introduces Tree-Path KL Divergence loss [25]. The idea is to encode the ground-truth path across all hierarchy levels into a single target distribution. Let 1_{y_l} be the one-hot vector of the ground-truth label at level l. The concatenated tree-path target is defined as

$$Y = \frac{1}{L}[1_{y_L}; \ldots; 1_{y_1}]. \tag{3.20}$$

The model outputs from all hierarchy levels are also concatenated and normalized using log-softmax:

$$\hat{Y} = \text{LogSoftmax}\left([f_L(Z_L^{class}); \ldots; f_1(Z_1^{class})]\right). \tag{3.21}$$

The Tree-Path KL loss is then computed as

$$\mathcal{L}_{TK} = \mathcal{L}_{KL}(\hat{Y}, Y). \tag{3.22}$$

This loss encourages the model to assign probability mass to a valid root-to-leaf path in the taxonomy. In the original H-CAST objective, the final training loss is

$$\mathcal{L}_{HCAST} = \mathcal{L}_{HV} + \alpha\mathcal{L}_{TK}, \tag{3.23}$$

where α controls the weight of the semantic consistency term.

3.3.4 Role of H-CAST in the Proposed Model

In this thesis, H-CAST serves as the image-only hierarchical backbone. Its role is to provide visually consistent hierarchical representations for order, family, and species prediction. The backbone preserves the fine-to-coarse visual grouping mechanism inherited from CAST and the multi-level supervision introduced by H-CAST.

The proposed method does not replace the H-CAST backbone. Instead, it extends the species-level branch with attribute-guided learning. This design is motivated by the observation that the species level is the most fine-grained level and is most likely to benefit from localized attribute cues. Family and order predictions are still produced from the original H-CAST hierarchy, while the species prediction is enhanced by an additional attribute-guided module.

Therefore, the H-CAST backbone provides two essential foundations for this thesis. First, it offers a structured visual hierarchy that aligns fine visual details with fine labels and coarser visual regions with coarser labels. Second, it provides the species-level graph-node features and class token used by the proposed attribute-guided module. The next sections describe how CUB attributes are represented, selected, and fused into this H-CAST-based framework.

3.4 Attribute Representation in CUB-200-2011

CUB-200-2011 provides rich visual attribute annotations in addition to species labels. The dataset contains 312 bird attributes that describe visual properties such as color,

shape, size, and pattern [32]. These attributes are especially useful for fine-grained bird recognition because many bird species differ only in subtle local traits, such as bill shape, crown color, wing pattern, belly color, or tail shape.

In this thesis, the CUB attributes are used as auxiliary supervision rather than as additional inputs during inference. Given an image, the model still predicts the taxonomic labels directly from the image. Attribute annotations are used only during training to guide the model toward discriminative bird traits.

3.4.1 Attribute Groups

In this thesis, the 312 CUB attributes are organized into 28 semantic groups based on their attribute names. Each group corresponds to a bird trait type, and attributes within the same group represent related states of the same visual property. For example, the group *bill shape* contains different bill-shape states, while the group *wing color* contains possible color states for the wing region. This group structure is important because attributes within the same group are mutually related and describe the same semantic property.

The attribute groups used in this thesis are shown in Table 3.1. The number of attribute states differs across groups. Color-related groups usually contain 15 states, while pattern-related groups usually contain fewer states.

Let the set of all attribute groups be denoted as

$$\mathcal{G} = \{g_1, g_2, \ldots, g_K\}, \tag{3.24}$$

where $K = 28$ for the full CUB attribute set. Each group g contains a set of attribute indices I_g. The full 312-dimensional attribute vector for a species c is denoted as

$$\mathbf{a}_c \in \mathbb{R}^{312}. \tag{3.25}$$

The class-level attribute vector is taken from the CUB class-level continuous attribute annotation file, where each entry indicates how frequently the attribute is considered present for a species.

For an image x_i with species label y_i^s, the corresponding attribute target is obtained from the class-level attribute annotation of that species:

$$\mathbf{a}_i = \mathbf{a}_{y_i^s}. \tag{3.26}$$

Table 3.1: CUB attribute groups used in this thesis. Each group contains a set of attribute states.

Attribute group	**# states**	**Attribute group**	**# states**
bill shape	9	wing color	15
upperparts color	15	underparts color	15
breast pattern	4	back color	15
tail shape	6	upper tail color	15
head pattern	11	breast color	15
throat color	15	eye color	14
bill length	3	forehead color	15
under tail color	15	nape color	15
belly color	15	wing shape	5
size	5	shape	14
back pattern	4	tail pattern	4
belly pattern	4	primary color	15
leg color	15	bill color	15
crown color	15	wing pattern	4

3.4.2 Group-wise Attribute Prediction

A straightforward design would be to predict the full 312-dimensional attribute vector directly. However, this thesis does not use that formulation. Instead, the model treats attribute prediction as a group-wise classification problem. This design follows the semantic organization of CUB attributes: attributes within the same group represent related states of the same visual property.

Although the original CUB annotations are provided as attribute-level presence labels, this thesis converts each group into a single-label classification target by selecting the highest-valued attribute state within the group.

For each attribute group g, the model predicts one attribute state from the states belonging to that group. The target for group g is obtained by selecting the attribute state with the highest value inside the group:

$$t_{i,g} = \arg\max_{j \in I_g} a_{i,j}. \tag{3.27}$$

Here, $t_{i,g}$ is the group-wise attribute target for sample i and group g.

The model uses a separate classifier for each active attribute group. Let $\mathbf{h}_{i,g}^{a}$ denote the attribute-oriented feature for group g. The group-specific classifier produces logits

$$\mathbf{z}_{i,g}^{a} = W_g \mathbf{h}_{i,g}^{a} + b_g, \tag{3.28}$$

where W_g and b_g are the parameters of the classifier for group g. The dimension of $\mathbf{z}_{i,g}^{a}$ is equal to the number of retained attribute states in that group.

This means that the model does not predict all CUB attributes using a single shared binary classifier. Instead, it predicts attribute states through multiple group-specific classifiers. This formulation is more consistent with the structure of CUB attributes because each classifier specializes in one semantic trait type, such as bill shape, wing color, or tail pattern.

3.4.3 Attribute Masks and Active Groups

When all 312 attributes are used, each group classifier predicts among all states in its corresponding group. However, this thesis also studies selected attribute subsets, such as HSDAS-199, Sibling-IVAS-188, and Random-199. In these settings, only a subset of attributes is retained.

Let $M_g \subseteq I_g$ denote the selected attribute indices for group g. If M_g is non-empty, then g is considered an active attribute group. The active group set is defined as

$$\mathcal{G}_{active} = \{g \in \mathcal{G} \mid |M_g| > 0\}. \tag{3.29}$$

For each active group, the original attribute indices are remapped to local class indices. If the target attribute state of a sample is retained in the selected mask, it is used as the target for the group-specific attribute loss. If the target attribute state is not retained, the attribute target for that sample and group is ignored during loss computation. This avoids forcing the model to predict an attribute state that is not included in the selected subset.

Formally, for a selected group g, the remapped target is

$$\tilde{t}_{i,g} = \phi_g(t_{i,g}), \tag{3.30}$$

where $\phi_g(\cdot)$ maps the original CUB attribute index to the local index inside M_g. If $t_{i,g} \notin M_g$, then $\tilde{t}_{i,g}$ is set to an ignore label and excluded from the attribute loss.

3.4.4 Role in the Proposed Model

The group-wise attribute representation serves two roles in the proposed method. First, it provides auxiliary supervision through group-specific attribute prediction heads. These heads encourage the model to learn features related to semantic bird traits. Second, the hidden attribute-oriented features are used for species-level fusion. Instead of using the predicted discrete attribute labels as inputs, the model uses continuous attribute-oriented representations learned before the group classifiers.

This distinction is important. The proposed model does not classify a bird by first predicting a complete attribute vector and then using that vector as the final representation. Rather, attribute prediction acts as an auxiliary task, while the learned attribute-oriented features are fused with the species-level visual representation. Therefore, the attribute module guides feature learning during training and supports species classification, while the inference process remains image-based.

Overall, this representation allows the model to exploit the semantic structure of CUB attributes without requiring manual attribute inputs at test time. It also enables different attribute selection strategies to be implemented through group masks, making it possible to compare FullAttr-312, HSDAS-199, Sibling-IVAS-188 and Random-199 under the same attribute-guided H-CAST framework.

3.5 Hard-Sibling Discriminative Attribute Selection

The previous section described how CUB attributes are represented as semantic groups and used as auxiliary supervision in the proposed model. However, CUB-200-2011 contains 312 attributes [32], and not all attributes are equally useful for distinguishing fine-grained bird species. Some attributes may be shared by many species, while others may be redundant or weakly discriminative. Therefore, this thesis studies an attribute selection strategy called *Hard-Sibling Discriminative Attribute Selection* (HSDAS).

Although the original selection implementation can compute attribute scores at the species, family, and order levels, the proposed model in this thesis uses the species-level selected attributes for attribute-guided fusion. This choice is consistent with the architecture described in this chapter, where the attribute-guided module is attached to the species-level branch of H-CAST [25]. Therefore, this section focuses only on species-level HSDAS.

3.5.1 Motivation

Fine-grained bird classification is challenging because visually similar species often belong to the same family. Such species may share global appearance, body shape, and dominant colors, but differ in localized visual traits such as bill shape, crown color, wing pattern, breast markings, or tail shape. A globally discriminative attribute may separate unrelated bird species, but may not help distinguish species that are taxonomically close. Therefore, the proposed selection strategy focuses on attributes that distinguish a species from its sibling species within the same family.

The key idea of HSDAS is to select attributes according to local taxonomic discrimination. For each species, the method compares its class-level attribute vector with the attribute vectors of sibling species. In addition, it gives special attention to hard siblings, which are sibling species that have similar attribute profiles. These hard siblings are likely to be difficult to distinguish, so attributes that separate them are expected to be useful for species-level recognition.

3.5.2 Species-Level Attribute Representation

Let the set of species be denoted as

$$\mathcal{Y}^s = \{1, 2, \ldots, C_s\}, \tag{3.31}$$

where $C_s = 200$ for CUB-200-2011. Each species c is associated with a class-level attribute vector

$$\mathbf{a}_c = [a_{c,1}, a_{c,2}, \ldots, a_{c,312}] \in \mathbb{R}^{312}. \tag{3.32}$$

In the implementation, the continuous CUB class-level attribute labels are normalized to the range $[0, 1]$. Each element $a_{c,j}$ indicates the class-level presence strength of attribute j for species c, derived from the continuous CUB attribute annotations.

The taxonomy defines a mapping from species to family:

$$\pi_f : \mathcal{Y}^s \to \mathcal{Y}^f. \tag{3.33}$$

For a target species c, its sibling species are defined as other species that share the same parent family:

$$\mathcal{S}(c) = \{c' \in \mathcal{Y}^s \mid \pi_f(c') = \pi_f(c),\ c' \neq c\}. \tag{3.34}$$

If a species has no sibling species, its attribute scores are set to zero and no attributes are selected from that species. In CUB, most species belong to families with multiple species, so sibling comparison is usually available.

3.5.3 General Sibling Discriminativeness

The first component of HSDAS measures how much stronger an attribute is for the target species compared with the average of its sibling species. For each species c and attribute j, the mean sibling attribute value is computed as

$$\bar{a}_{\mathcal{S}(c),j} = \frac{1}{|\mathcal{S}(c)|} \sum_{c' \in \mathcal{S}(c)} a_{c',j}. \tag{3.35}$$

The general sibling discriminativeness score is then defined as

$$d_{\text{all}}(c, j) = \max\left(a_{c,j} - \bar{a}_{\mathcal{S}(c),j}, 0\right). \tag{3.36}$$

This score keeps only positive differences. If an attribute is not stronger for the target species than for its siblings on average, it is not considered discriminative for that species. This design makes the selected attributes species-specific: an attribute is useful only when it helps distinguish the target species from its local taxonomic neighborhood. Additionally, this formulation focuses on positive presence cues so attributes that are mainly discriminative through absence are not selected by this scoring rule.

3.5.4 Hard-Sibling Discriminativeness

The second component focuses on hard sibling species. Hard siblings are defined as sibling species whose attribute vectors are most similar to the target species. The similarity between species c and a sibling species c' is computed using cosine similarity:

$$\text{sim}(c, c') = \frac{\mathbf{a}_c^\top \mathbf{a}_{c'}}{\|\mathbf{a}_c\|_2 \|\mathbf{a}_{c'}\|_2}. \tag{3.37}$$

For each species c, the hard sibling set $\mathcal{H}_r(c)$ contains the top r sibling species with the highest cosine similarity:

$$\mathcal{H}_r(c) \subseteq \mathcal{S}(c). \tag{3.38}$$

In the experiments of this thesis, the species-level setting uses $r = 3$.

For each hard sibling $h \in \mathcal{H}_r(c)$, the positive attribute difference is computed as

$$\Delta(c, h, j) = \max(a_{c,j} - a_{h,j}, 0). \tag{3.39}$$

The hard-sibling discriminativeness score is a similarity-weighted sum of these positive differences:

$$d_{\text{hard}}(c, j) = \sum_{h \in \mathcal{H}_r(c)} w_{c,h} \Delta(c, h, j), \tag{3.40}$$

where the weight $w_{c,h}$ is the normalized cosine similarity:

$$w_{c,h} = \frac{\text{sim}(c, h)}{\sum_{h' \in \mathcal{H}_r(c)} \text{sim}(c, h') + \epsilon}. \tag{3.41}$$

This weighting gives more influence to sibling species that are more similar to the target species. As a result, the method emphasizes attributes that separate a species from its most confusing relatives.

3.5.5 Final Attribute Score

The final HSDAS score combines the general sibling discriminativeness and the hard-sibling discriminativeness:

$$s(c, j) = d_{\text{all}}(c, j) + d_{\text{hard}}(c, j). \tag{3.42}$$

The first term captures attributes that distinguish the target species from all sibling species on average. The second term captures attributes that distinguish the target species from the most similar sibling species. Combining both terms allows the method to balance general local discrimination and hard-case discrimination.

For each species, attributes with zero final score are removed from consideration. The remaining attributes are sorted in descending order by three criteria:

$$s(c, j), \quad d_{\text{hard}}(c, j), \quad d_{\text{all}}(c, j). \tag{3.43}$$

The final score is the primary ranking criterion. The hard-sibling score is used as the first tie-breaker, giving priority to attributes that help with difficult sibling distinctions. The general sibling score is used as the second tie-breaker.

3.5.6 Top-k Selection and Attribute Union

After scoring all attributes for each species, HSDAS selects the top k attributes for each species. In this thesis, the species-level setting uses $k = 5$. Let $\mathcal{T}_k(c)$ denote the top-k selected attributes for species c:

$$\mathcal{T}_k(c) = \mathrm{TopK}_k\left(\{j \mid s(c,j) > 0\},\ s(c,j)\right). \tag{3.44}$$

The final selected attribute set is obtained by taking the union of the selected attributes over all species:

$$\mathcal{M} = \bigcup_{c \in \mathcal{Y}^s} \mathcal{T}_k(c). \tag{3.45}$$

This union operation produces a compact global attribute subset for the dataset. Although attributes are selected based on species-level sibling discrimination, the final mask is shared by the attribute-guided model. In the main setting of this thesis, this process produces the HSDAS-199 attribute subset.

The selected attributes are then converted into a group mask according to the 28 CUB attribute groups. For each group g, the mask contains only the selected attribute indices that belong to that group:

$$M_g = \mathcal{M} \cap I_g, \tag{3.46}$$

where I_g is the full index set of attributes in group g. These group masks define the active attribute states used by the group-specific attribute classifiers.

3.5.7 Algorithm Summary

Algorithm 1 summarizes the species-level HSDAS procedure used in this thesis. The detailed definitions of d_{all}, d_{hard}, and $s(c,j)$ are given in the previous subsections.

3.5.8 Use in the Proposed Model

The selected HSDAS attribute set is used to build the attribute mask for the proposed attribute-guided H-CAST model. This mask determines which attribute states are retained in each attribute group and therefore controls the output dimensions of the group-specific attribute classifiers.

Importantly, HSDAS is used only to construct the training supervision mask. It does not introduce additional inputs during inference. At test time, the model still receives

Algorithm 1 Species-Level Hard-Sibling Discriminative Attribute Selection

Require: Attribute matrix $A \in \mathbb{R}^{C_s \times 312}$, species-to-family mapping π_f, number of hard siblings r, number of selected attributes per species k

Ensure: Selected attribute set $\mathcal{M}$

Initialize $\mathcal{M} \leftarrow \emptyset$
for each species $c \in \mathcal{Y}^s$ **do**
  Find sibling species: $\mathcal{S}(c) \leftarrow \{c' \in \mathcal{Y}^s \mid \pi_f(c') = \pi_f(c), c' \neq c\}$
  if $\mathcal{S}(c) = \emptyset$ **then**
    Continue
  end if
  Compute mean sibling attribute vector: $\bar{\mathbf{a}}_{\mathcal{S}(c)} \leftarrow |\mathcal{S}(c)|^{-1} \sum_{c' \in \mathcal{S}(c)} \mathbf{a}_{c'}$
  Compute $d_{\text{all}}(c, j)$ for all attributes $j = 1, \ldots, 312$
  Compute cosine similarity between c and each sibling $c' \in \mathcal{S}(c)$
  Select hard siblings $\mathcal{H}_r(c)$ as the top-r most similar species in $\mathcal{S}(c)$
  Compute normalized hard-sibling weights $w_{c,h}$ for all $h \in \mathcal{H}_r(c)$
  Compute $d_{\text{hard}}(c, j)$ and final score $s(c, j) = d_{\text{all}}(c, j) + d_{\text{hard}}(c, j)$ for all j
  Select top-k positive-score attributes:

$$\mathcal{T}_k(c) \leftarrow \text{TopK}_k\left(\{j \mid s(c, j) > 0\}, s(c, j)\right)$$

  Update selected set: $\mathcal{M} \leftarrow \mathcal{M} \cup \mathcal{T}_k(c)$
end for
return $\mathcal{M}$

only an image. The selected attributes influence the model indirectly by supervising the attribute-guided module during training and by shaping the attribute-oriented features used for species-level fusion.

This design has two advantages. First, it reduces the attribute space from the full 312 attributes to a compact subset focused on species-level discrimination. Second, it aligns attribute selection with the fine-grained nature of the task by prioritizing attributes that distinguish sibling and hard-sibling species. Therefore, HSDAS provides a taxonomy-aware attribute subset that is more directly connected to the objective of species-level bird classification than a purely global attribute selection criterion.

3.6 Attribute-Guided Learning Module

This section describes the attribute-guided learning module added on top of the H-CAST backbone [25]. The purpose of this module is to use CUB attribute annotations [32]

as auxiliary supervision and to inject attribute-oriented information into species-level prediction. The module is designed specifically for the fine-grained species branch because species recognition requires subtle local cues, such as bill shape, wing pattern, breast color, crown color, and tail shape.

The proposed module has four main components: attribute-group queries, cross-attention over species-level graph features, group-specific attribute prediction heads, and species-level attribute fusion. Importantly, the module does not use one query or one classifier for each individual CUB attribute. Instead, it operates at the level of active attribute groups. Each active group corresponds to a semantic attribute type, such as *bill shape*, *wing color*, *breast pattern*, or *tail shape*.

3.6.1 Attribute-Group Queries

Let the set of active attribute groups be denoted as

$$\mathcal{G}_{active} = \{g_1, g_2, \ldots, g_G\}, \tag{3.47}$$

where G is the number of active groups after applying the attribute mask. For the FullAttr-312 setting, $G = 28$ but for selected attribute subsets, groups with no retained attributes are removed.

The model maintains one learnable query for each active attribute group. Let the learnable attribute-query matrix be

$$Q^a = [\mathbf{q}_{g_1}, \mathbf{q}_{g_2}, \ldots, \mathbf{q}_{g_G}] \in \mathbb{R}^{G \times D}, \tag{3.48}$$

where D is the hidden dimension of the H-CAST backbone. Each query $\mathbf{q}_g$ is associated with one semantic attribute group g, not with one individual attribute. Therefore, the number of attribute queries is equal to the number of active attribute groups, rather than the total number of selected attributes.

For a mini-batch of size B, the query matrix is expanded to

$$Q_B^a \in \mathbb{R}^{B \times G \times D}. \tag{3.49}$$

Before being used in attention, the queries are normalized with a layer normalization module. This normalization helps align the scale of the attribute queries with the H-CAST feature representations.

3.6.2 Cross-Attention over Species-Level Graph Features

The attribute queries attend to graph-node features from the species-level stage of H-CAST. Let

$$H^s \in \mathbb{R}^{B\times N_s\times D} \tag{3.50}$$

denote the species-level graph-node features, where N_s is the number of graph nodes at the species-level stage. Let

$$\mathbf{c}^s \in \mathbb{R}^{B\times D} \tag{3.51}$$

denote the species-level class token.

For the CUB three-level hierarchy, attribute-group queries attend to graph-node features from the species-level stage. In the implementation, when the model predicts three levels, the species-level branch uses the graph nodes from the second H-CAST block and the corresponding species class token. The family and order predictions are produced from later, coarser H-CAST stages.

The cross-attention operation uses attribute-group queries as queries, and species-level graph-node features as keys and values:

$$H^a = \mathrm{MHA}\left(Q_B^a, H^s, H^s\right), \tag{3.52}$$

where $\mathrm{MHA}(\cdot)$ denotes multi-head attention [31]. The output is

$$H^a = [\mathbf{h}_{g_1}^a, \mathbf{h}_{g_2}^a, \ldots, \mathbf{h}_{g_G}^a] \in \mathbb{R}^{B\times G\times D}. \tag{3.53}$$

Each vector $\mathbf{h}_g^a$ is an attribute-oriented representation for group g. Intuitively, this operation allows each attribute group to extract visual evidence from the species-level graph nodes. For example, a query corresponding to *bill shape* may learn to attend to regions related to the bird's bill, while a query corresponding to *wing color* may learn to focus on wing-related segments. However, the model is not explicitly supervised with pixel-level attribute localization, so these attention patterns should be interpreted as learned internal representations rather than guaranteed attribute localizations.

A key-padding mask is used in the attention operation to ignore invalid graph nodes introduced by padding. After cross-attention, the attribute-oriented features are normalized and passed through dropout:

$$H^a = \mathrm{Dropout}\left(\mathrm{LN}(H^a)\right). \tag{3.54}$$

3.6.3 Group-Specific Attribute Prediction

The attribute-oriented features are supervised by group-specific attribute prediction heads. For each active group $g \in \mathcal{G}_{active}$, the model defines a separate linear classifier:

$$f_g^a : \mathbb{R}^D \to \mathbb{R}^{C_g}, \tag{3.55}$$

where C_g is the number of retained attribute states in group g. The classifier for group g receives the corresponding attribute-oriented feature $\mathbf{h}_g^a$ and produces logits:

$$\mathbf{z}_g^a = f_g^a \left(\mathrm{LN}(\mathbf{h}_g^a) \right). \tag{3.56}$$

This design means that the model does not predict a single 312-dimensional binary attribute vector. Instead, it predicts one retained attribute category for each active attribute group. For example, the classifier for *bill shape* predicts among the retained bill-shape states, while the classifier for *wing color* predicts among the retained wing-color states.

The group-specific attribute predictions are used only as auxiliary outputs during training. Let $\tilde{t}_{i,g}$ be the remapped target label for sample i and group g. Let Ω_g denote the set of samples whose remapped target for group g is valid, i.e., not ignored by the selected mask. The set of valid groups in a mini-batch is

$$\mathcal{G}_{valid} = \{ g \in \mathcal{G}_{active} \mid |\Omega_g| > 0 \}. \tag{3.57}$$

The attribute prediction loss is computed as

$$\mathcal{L}_{attr} = \frac{1}{|\mathcal{G}_{valid}|} \sum_{g \in \mathcal{G}_{valid}} \frac{1}{|\Omega_g|} \sum_{i \in \Omega_g} \mathcal{L}_{CE} \left(\mathbf{z}_{i,g}^a, \tilde{t}_{i,g} \right) \tag{3.58}$$

The role of this auxiliary attribute loss is to encourage each group query to learn features related to its corresponding semantic trait. These attribute predictions are not used as discrete inputs for species classification. Instead, the continuous attribute-oriented features are used in the fusion stage.

3.6.4 Species-Level Attribute Fusion

After obtaining the attribute-oriented features H^a, the model aggregates them into a single attribute summary vector. This is done using attention pooling conditioned on

the species-level class token. The species class token is used as the query, while the attribute-oriented features are used as keys and values:

$$\tilde{\mathbf{h}}^{a}_{sum} = \mathrm{MHA}\left(\mathbf{c}^{s}_{1}, H^{a}, H^{a}\right), \qquad \mathbf{c}^{s}_{1} \in \mathbb{R}^{B\times 1\times D}, \tag{3.59}$$

where $\mathbf{c}^{s}_{1}$ is the species class token with an added sequence dimension. The output is then squeezed to obtain

$$\mathbf{h}^{a}_{sum} \in \mathbb{R}^{B\times D}. \tag{3.60}$$

This attention-pooling step allows the species-level visual representation to select and combine useful attribute-group features. Instead of treating all attribute groups equally, the model can assign different importance to different attribute-oriented representations depending on the input image.

The resulting attribute summary is concatenated with the species-level class token:

$$\mathbf{h}^{fuse}_{in} = [\mathbf{c}^{s}; \mathbf{h}^{a}_{sum}] \in \mathbb{R}^{B\times 2D}. \tag{3.61}$$

The concatenated vector is then passed through a fusion multilayer perceptron:

$$\mathbf{h}^{s}_{fused} = \mathrm{MLP}_{fuse}\left(\mathbf{h}^{fuse}_{in}\right), \tag{3.62}$$

where the fusion module maps the $2D$-dimensional concatenated representation back to the original hidden dimension D. In the implementation, this fusion module consists of a linear layer, a GELU activation, and layer normalization.

The final species classifier uses the fused species representation:

$$\mathbf{z}^{s} = f_{s}\left(\mathbf{h}^{s}_{fused}\right). \tag{3.63}$$

The attribute-guided fusion is applied to the species-level branch. Family and order predictions are still produced from the original H-CAST hierarchical stages. In the three-level CUB setting, the family classifier uses the family-level H-CAST class token, and the order classifier uses the coarser order-level H-CAST class token:

$$\mathbf{z}^{f} = f_{f}(\mathbf{c}^{f}), \qquad \mathbf{z}^{o} = f_{o}(\mathbf{c}^{o}). \tag{3.64}$$

Therefore, attributes do not directly modify the family or order classifiers. Their primary effect is on species-level classification, where fine-grained attribute cues are expected to be most useful. Family and order predictions may still be indirectly affected during training because the whole network is optimized jointly, but there is no direct attribute fusion into those branches.

3.6.5 Summary

The attribute-guided learning module extends H-CAST by adding attribute-oriented supervision and species-level fusion. It uses one learnable query for each active attribute group, applies cross-attention over species-level graph-node features, predicts retained attribute categories with group-specific classifiers, and fuses the attention-pooled attribute summary with the species class token. This design allows the model to exploit CUB attributes during training while keeping inference image-based. At test time, no manually provided attribute labels are required; the attribute queries and fusion module operate internally using only the image-derived H-CAST features.

3.7 Training Objective

The proposed model is trained with a combination of hierarchical visual consistency loss, optional Tree-Path KL divergence loss, auxiliary attribute prediction loss, and attribute attention diversity loss. The overall objective is designed to preserve the original hierarchical classification ability of H-CAST [25], while encouraging the species-level representation to encode discriminative attribute-oriented information.

For the three-level CUB hierarchy, the model predicts species, family, and order labels. In addition, it predicts group-wise attribute labels for the active attribute groups selected by the attribute mask. The total training objective is written as

$$\mathcal{L} = \mathcal{L}_{HV} + \alpha\mathcal{L}_{TK} + \lambda_{attr}\mathcal{L}_{attr} + \lambda_{div}\mathcal{L}_{div}, \tag{3.65}$$

where $\mathcal{L}_{HV}$ is the hierarchical visual consistency loss inherited from H-CAST, $\mathcal{L}_{TK}$ is the optional Tree-Path KL divergence loss inherited from H-CAST, $\mathcal{L}_{attr}$ is the auxiliary attribute prediction loss, and $\mathcal{L}_{div}$ is the attribute attention diversity loss. The weights α, λ_{attr}, and λ_{div} control the contribution of each additional loss term.

In the original H-CAST formulation, $\mathcal{L}_{HV}$ denotes the sum of classification losses across hierarchy levels. In this thesis, it corresponds to the level-wise classification loss over species, family, and order. In the implementation, $\mathcal{L}_{TK}$ is enabled only when the global KL option is used. If this option is disabled, α is effectively set to zero. Similarly, setting $\lambda_{div} = 0$ removes the diversity regularization term, and setting $\lambda_{attr} = 0$ removes the attribute prediction supervision.

3.7.1 Hierarchical Visual Consistency Loss

The main task is hierarchical classification over species, family, and order. Let $\mathbf{z}^s$, $\mathbf{z}^f$, and $\mathbf{z}^o$ denote the logits predicted by the model for species, family, and order, respectively. Let y^s, y^f, and y^o denote the corresponding ground-truth labels.

Following H-CAST, the hierarchical visual consistency loss is defined as the sum of classification losses at the three hierarchy levels:

$$\mathcal{L}_{HV} = \mathcal{L}_s + \mathcal{L}_f + \mathcal{L}_o, \tag{3.66}$$

where

$$\mathcal{L}_s = \mathcal{L}_{CE}(\mathbf{z}^s, y^s), \qquad \mathcal{L}_f = \mathcal{L}_{CE}(\mathbf{z}^f, y^f), \qquad \mathcal{L}_o = \mathcal{L}_{CE}(\mathbf{z}^o, y^o). \tag{3.67}$$

In H-CAST, different hierarchy levels are supervised at different stages of the visual hierarchy. Fine-grained species prediction is associated with a more detailed species-level representation, while family and order predictions are produced from later, coarser H-CAST stages. In the proposed model, the species-level representation is further enhanced by attribute-guided fusion, whereas family and order predictions remain produced by the original H-CAST hierarchical stages.

In practice, the base classification criterion may be standard cross-entropy, label-smoothed cross-entropy, or soft-target cross-entropy when Mixup [40] or CutMix [39] is used. Regardless of the exact criterion, $\mathcal{L}_{HV}$ supervises all three taxonomic levels jointly. This follows the H-CAST principle that different hierarchy levels should be trained together within a shared visually grounded hierarchy rather than using separate independent classifiers for each level.

3.7.2 Attribute Prediction Loss

The attribute branch is supervised through group-specific attribute prediction heads. Let $\mathcal{G}_{active}$ denote the set of active attribute groups after applying the selected attribute mask. For each active group $g \in \mathcal{G}_{active}$, the model predicts a logit vector $\mathbf{z}_g^a$ over the retained attribute states in that group.

As described in the previous section, the original CUB attribute targets are remapped according to the selected group mask. Let $\tilde{t}_{i,g}$ be the remapped target for sample i and group g. If the original ground-truth attribute state is not retained in the selected mask, then $\tilde{t}_{i,g}$ is assigned an ignore label and excluded from the loss.

Let Ω_g denote the set of samples in the mini-batch whose target for group g is not ignored. The set of valid groups is

$$\mathcal{G}_{valid} = \{g \in \mathcal{G}_{active} \mid |\Omega_g| > 0\}. \tag{3.68}$$

The attribute prediction loss is computed as

$$\mathcal{L}_{attr} = \frac{1}{|\mathcal{G}_{valid}|} \sum_{g \in \mathcal{G}_{valid}} \frac{1}{|\Omega_g|} \sum_{i \in \Omega_g} \mathcal{L}_{CE}\left(\mathbf{z}^a_{i,g}, \tilde{t}_{i,g}\right). \tag{3.69}$$

This auxiliary loss encourages each attribute-group query to learn a representation related to its semantic group, such as bill shape, wing color, head pattern, or tail shape. The attribute loss is not used to make the final prediction directly. Instead, it shapes the attribute-oriented features that are later fused with the species-level class token. During inference, no ground-truth attribute labels are required.

3.7.3 Tree-Path KL Divergence Loss

H-CAST introduces a semantic consistency objective to reduce invalid prediction paths in the taxonomy [25]. In this thesis, this term is inherited as an optional global KL loss. It is used only when the corresponding training option is enabled.

Let the logits from the three hierarchy levels be concatenated as

$$\mathbf{z}^{all} = [\mathbf{z}^o; \mathbf{z}^f; \mathbf{z}^s]. \tag{3.70}$$

The corresponding target vectors are also concatenated:

$$\mathbf{y}^{all} = [\mathbf{y}^o; \mathbf{y}^f; \mathbf{y}^s], \tag{3.71}$$

where the target vectors are represented as one-hot vectors, or as soft target vectors when Mixup or CutMix is applied. The concatenated target vector is normalized so that it forms a valid distribution:

$$\bar{\mathbf{y}}^{all} = \frac{\mathbf{y}^{all}}{\|\mathbf{y}^{all}\|_1}. \tag{3.72}$$

The Tree-Path KL loss is then computed as

$$\mathcal{L}_{TK} = \mathcal{L}_{KL}\left(\log \mathrm{Softmax}(\mathbf{z}^{all}), \bar{\mathbf{y}}^{all}\right). \tag{3.73}$$

This loss encourages the concatenated predictions across order, family, and species to align with the ground-truth taxonomic path. In the implementation, this term is weighted by α, corresponding to the global KL weight. If the global KL option is not used, this term is omitted from the total loss.

3.7.4 Attribute Attention Diversity Loss

The proposed model uses multiple attribute-group queries to extract attribute-oriented features from species-level graph nodes. Without additional regularization, different attribute groups may attend to similar graph regions, which can make the attribute representations redundant. To reduce this issue, this thesis studies an attribute attention diversity loss.

The diversity loss is inspired by the diversity component of Mutual-Channel Loss, but is adapted to attribute-group attention maps rather than class-specific convolutional channels [4]. In the original Mutual-Channel Loss, the diversity component encourages different channels to focus on different discriminative regions. In this thesis, the same high-level idea is applied to attribute-group attention: different semantic attribute groups should be encouraged to attend to complementary graph nodes.

Let

$$A \in \mathbb{R}^{B \times G \times N_s} \tag{3.74}$$

denote the cross-attention weights from the attribute-group queries to the species-level graph nodes, where B is the batch size, G is the number of active attribute groups, and N_s is the number of species-level graph nodes. Although the cross-attention module returns attention weights, the implementation applies an additional softmax normalization along the graph-node dimension before computing the diversity score:

$$\tilde{A}_{b,g,n} = \mathrm{Softmax}_n(A_{b,g,n}). \tag{3.75}$$

Instead of computing diversity over all individual attribute groups directly, the implementation first groups active attribute groups into broader macro-parts which are head, wing, tail, and body. Let $\mathcal{P}$ denote the set of active macro-parts. For each macro-part $p \in \mathcal{P}$, the attention map is computed by averaging the attention maps of all active attribute groups belonging to that macro-part:

$$M_{b,p,n} = \frac{1}{|\mathcal{G}_p|} \sum_{g \in \mathcal{G}_p} \tilde{A}_{b,g,n}, \tag{3.76}$$

where $\mathcal{G}_p$ is the set of active attribute groups assigned to macro-part p.

The diversity score for each sample is defined as

$$h_b = \sum_{n=1}^{N_s} \max_{p \in \mathcal{P}} M_{b,p,n}. \tag{3.77}$$

This score becomes larger when different macro-parts cover complementary graph nodes. If all macro-parts attend to the same region, the maximum attention map does not cover many distinct nodes and the score remains lower. Therefore, maximizing h_b encourages different attribute macro-parts to distribute their attention over different visual evidence.

The diversity loss is defined as the negative mean diversity score:

$$\mathcal{L}_{div} = -\frac{1}{B}\sum_{b=1}^{B} h_b. \tag{3.78}$$

Since the training objective is minimized, adding $\lambda_{div}\mathcal{L}_{div}$ encourages the model to maximize the diversity score. If fewer than two macro-parts are active, the diversity loss is not meaningful and is set to zero.

This loss does not use any pixel-level attribute localization labels. It only regularizes the internal cross-attention patterns learned by the attribute-group queries. Therefore, it should be understood as a soft diversity regularizer rather than direct supervision for attribute localization.

3.7.5 Overall Objective

Combining the above components, the final training loss is

$$\mathcal{L} = \mathcal{L}_{HV} + \alpha\mathcal{L}_{TK} + \lambda_{attr}\mathcal{L}_{attr} + \lambda_{div}\mathcal{L}_{div}. \tag{3.79}$$

In the experiments, different settings can be obtained by changing the loss weights. For example, setting $\lambda_{div} = 0$ removes the attention diversity regularization, while setting $\lambda_{div} = 0.5$ enables the diversity term. Similarly, the contribution of attribute supervision is controlled by λ_{attr}, and the semantic consistency loss is controlled by α.

This objective combines hierarchical label supervision, semantic consistency regularization, auxiliary attribute supervision, and attention diversity regularization. The hierarchical classification loss trains the model to predict species, family, and order. The Tree-Path KL term encourages semantic consistency across taxonomy levels. The attribute prediction loss encourages the species-level branch to encode semantic bird traits. The diversity loss encourages attribute macro-parts to attend to complementary visual regions. Together, these components aim to improve fine-grained species discrimination while preserving hierarchical classification behavior.

3.8 Inference Procedure

Although the attribute branch is trained with attribute labels, the inference process remains image-only. At test time, the model receives only an input image. The initial superpixel segmentation required by the H-CAST/CAST backbone [14, 25] is generated automatically from the image and is not manually provided.

Given an input image x, the model first computes hierarchical visual representations using the H-CAST backbone. The backbone produces class-token and graph-node features at different stages of the visual hierarchy. The family and order predictions are obtained from the corresponding H-CAST stages, following the original hierarchical classification pipeline.

For species prediction, the model additionally uses the attribute-guided module. The attribute-group queries attend to the species-level graph-node features and produce attribute-oriented representations internally. These representations are not compared with ground-truth attribute labels during inference, since attribute labels are not used at test time. Instead, they are used only as learned visual features.

The attribute-oriented features are aggregated by the attention-pooling module to obtain an attribute summary vector. This summary vector is concatenated with the species-level class token, and the concatenated representation is passed through the fusion MLP to obtain the fused species feature:

$$\mathbf{h}^s_{fused} = \mathrm{MLP}_{fuse}\left([\mathbf{c}^s; \mathbf{h}^a_{sum}]\right). \tag{3.80}$$

The final species logits are then computed from the fused species feature:

$$\mathbf{z}^s = f_s(\mathbf{h}^s_{fused}). \tag{3.81}$$

The family and order logits are computed from the original H-CAST hierarchical stages:

$$\mathbf{z}^f = f_f(\mathbf{c}^f), \qquad \mathbf{z}^o = f_o(\mathbf{c}^o). \tag{3.82}$$

The predicted labels are obtained by taking the class with the highest logit value at each level:

$$\hat{y}^s = \arg\max_{c \in \mathcal{Y}^s} z^s_c, \qquad \hat{y}^f = \arg\max_{c \in \mathcal{Y}^f} z^f_c, \qquad \hat{y}^o = \arg\max_{c \in \mathcal{Y}^o} z^o_c. \tag{3.83}$$

Therefore, the model produces predictions for all three taxonomy levels:

$$(\hat{y}^s, \hat{y}^f, \hat{y}^o), \tag{3.84}$$

following the notation used in the problem formulation. When reported as a root-to-leaf taxonomy path, the same prediction can also be written as

$$(\hat{y}^o, \hat{y}^f, \hat{y}^s), \tag{3.85}$$

corresponding to order, family, and species. The attribute branch contributes to the prediction only through its learned internal representations. It does not require attribute annotations, manual attribute descriptions, or any additional non-image input during inference.

This design preserves the practical inference setting of the original H-CAST model. The proposed attribute-guided module improves representation learning during training, but deployment remains the same as a standard image-based hierarchical classifier.

3.9 Discussion

The proposed method is designed to address the main limitation of the image-only H-CAST baseline in fine-grained bird classification. H-CAST provides a strong hierarchical visual backbone by organizing image regions from fine to coarse and producing predictions at species, family, and order levels [25]. However, it does not explicitly use the attribute annotations available in CUB-200-2011 [32]. Since many bird species differ by subtle visual traits, this thesis introduces attribute-guided learning to encourage the species-level representation to encode discriminative bird characteristics.

3.9.1 Rationale of the Proposed Design

Species-level bird classification requires fine-grained local evidence. Two species from the same family may have similar global appearance, but differ in localized traits such as bill shape, wing pattern, crown color, breast markings, or tail shape. These traits are directly reflected in the CUB attribute annotations. Therefore, using attributes as auxiliary supervision is a natural way to guide the model toward visual cues that are relevant for species discrimination.

The proposed model attaches the attribute-guided module to the species-level branch because the species level is the finest and most difficult level in the hierarchy. Family and order classification are coarser tasks and may rely more on broader visual structures. In contrast, species recognition often depends on subtle part-level differences. For this

reason, the attribute-guided fusion is applied to the species-level branch, while family and order predictions remain produced by the original H-CAST hierarchical stages.

The use of attribute-group queries is also motivated by the structure of the CUB attributes. Instead of predicting 312 independent binary attributes, the model operates on semantic attribute groups such as *bill shape*, *wing color*, *head pattern*, and *tail shape*. This group-wise formulation is more aligned with the organization of CUB attributes and allows each query to specialize in one semantic trait type. The group-specific attribute classifiers then provide auxiliary supervision for these attribute-oriented representations.

HSDAS further improves the attribute usage strategy by selecting attributes according to species-level sibling discrimination. Using all 312 attributes may introduce redundancy or weakly useful supervision. HSDAS reduces this issue by prioritizing attributes that distinguish a species from its sibling species, especially hard siblings with similar attribute profiles. This makes the selected attribute subset more aligned with the fine-grained nature of the classification task.

The attribute attention diversity loss is introduced to reduce redundancy among attribute-related attention patterns. Since different attribute groups may correspond to different bird parts or visual properties, they should ideally capture complementary evidence. The diversity regularizer encourages macro-part attention maps to cover different graph nodes, helping the model avoid relying on only one dominant region.

3.9.2 Advantages

The first advantage of the proposed method is that it does not require attribute labels during inference. Attribute annotations are used only during training to supervise the attribute branch. At test time, the model receives only an image and produces order, family, and species predictions. Therefore, the inference setting remains the same as the original image-only H-CAST model.

Second, the method is compatible with the official CUB-200-2011 train/test protocol. The model can be trained using the standard image labels and class-level attribute annotations provided by the dataset. No additional manual annotation, part localization label, or attribute input is required.

Third, the group-wise attribute design improves interpretability. Since each attribute query corresponds to a semantic group, the learned attention patterns can be analyzed

according to bird traits such as head, wing, tail, and body-related properties. Although these attention maps are not guaranteed to be precise attribute localization maps, they provide useful qualitative evidence about how the model uses attribute-oriented information.

Fourth, HSDAS provides a more compact attribute subset than FullAttr-312. By selecting attributes that are useful for distinguishing sibling species, HSDAS is designed to reduce the attribute space while preserving discriminative semantic cues. This helps avoid using all attributes indiscriminately and provides a more task-specific attribute supervision mask.

Finally, the diversity loss encourages different attribute macro-parts to attend to complementary graph-node regions. This is useful because bird recognition often requires multiple cues from different parts of the body. Encouraging diversity among attribute attention maps is intended to reduce redundancy and make the attribute-guided representation more informative.

3.9.3 Limitations

Despite these advantages, the proposed method also has several limitations. First, it depends on the availability and quality of class-level attribute annotations. The approach is suitable for datasets such as CUB-200-2011, where attribute annotations are provided. However, it may not be directly applicable to datasets without reliable semantic attributes. Another limitation is that the attribute targets used in this thesis are class-level attributes. Therefore, all images from the same species share the same attribute target, even though the visibility of a trait may vary across images due to pose, occlusion, viewpoint, or image quality. This makes the attribute supervision suitable for injecting species-level semantic priors, but less capable of modeling image-specific attribute presence.

Second, HSDAS is designed for settings where both a taxonomy and class-level attributes are available. The method relies on sibling relationships in the taxonomy and compares class-level attribute profiles to identify hard siblings. Therefore, its usefulness may be limited in datasets without a meaningful hierarchy or without class-level semantic descriptions.

Third, the current attribute fusion is applied mainly to the species-level branch. This design is reasonable for fine-grained bird recognition, but it means that family and order

classifiers do not directly receive fused attribute representations. Future work could study whether attribute guidance at coarser levels can further improve hierarchical consistency.

Fourth, the method does not use pixel-level attribute localization supervision. Although attribute-group queries attend to species-level graph nodes, the model is not explicitly trained to localize attributes in the image. Therefore, attention visualizations should be interpreted as internal attention patterns rather than ground-truth explanations of where each attribute appears.

Finally, the performance of the method may depend on the selected attribute mask and the loss weights. Different choices of λ_{attr}, λ_{div}, and the selected attribute subset can affect the balance between classification, attribute prediction, and attention diversity. This makes ablation studies necessary to understand the contribution of each component.

Overall, the proposed method extends H-CAST in a targeted way. It preserves the visually consistent hierarchical backbone, adds attribute-guided learning where fine-grained discrimination is most needed, and uses HSDAS to select a compact set of discriminative attributes. The method is therefore well suited to the CUB-200-2011 setting, where species recognition depends on subtle bird traits and where both taxonomy and attribute annotations are available.

Chapter 4

EXPERIMENTS AND EVALUATION

4.1 Research Questions

The experiments in this chapter are designed to answer the following research questions:

- **RQ1:** Does the proposed attribute-guided H-CAST improve hierarchical bird classification compared with the image-only H-CAST baseline under the official CUB-200-2011 train/test protocol?
- **RQ2:** Does using a compact selected attribute subset preserve or improve performance compared with using all 312 CUB attributes?
- **RQ3:** Does the proposed HSDAS strategy outperform equally sized random subsets and the internal Sibling-IVAS statistical baseline?
- **RQ4:** How do the proposed attribute-guided components, including species-level fusion, auxiliary attribute prediction, and diversity regularization, affect species accuracy, Full-Path Accuracy, wAP, TICE, and attribute prediction accuracy?

These questions correspond directly to the main goals of this thesis. RQ1 evaluates whether attribute-guided learning improves the image-only hierarchical baseline. RQ2 studies whether the proposed hard-sibling-aware attribute selection is more effective than using all attributes, random subsets, or earlier internal selection baselines that do not explicitly model hard siblings. RQ3 analyzes the role of the diversity regularization term. RQ4 examines how the attribute branch contributes to the final prediction behavior, including both classification performance and auxiliary attribute prediction quality.

4.2 Dataset and Experimental Protocol

This section describes the dataset, taxonomy structure, attribute annotations, train/test protocol, and implementation details used in the experiments. The main experiments are conducted on the CUB-200-2011 dataset and follow the official train/test split used in the

H-CAST benchmark. This choice ensures that the results are directly comparable with the image-only H-CAST baseline and the reported results of the original H-CAST paper.

4.2.1 CUB-200-2011

The experiments are conducted on the Caltech-UCSD Birds-200-2011 dataset, commonly referred to as CUB-200-2011 [32]. CUB-200-2011 is a standard benchmark for fine-grained bird image classification. It contains 200 bird species and 11,788 images. The official split consists of 5,994 training images and 5,794 test images.

CUB-200-2011 is suitable for this thesis for three reasons. First, it is a fine-grained classification dataset where many classes are visually similar and differ only in subtle local traits. Second, the dataset provides rich attribute annotations describing bird appearance, which makes it possible to study attribute-guided representation learning. Third, the bird species can be organized into a biological taxonomy, allowing the task to be formulated as hierarchical classification rather than flat species classification.

In this thesis, each input image is associated with three taxonomic labels: order, family, and species. The model is trained to predict all three levels. The species label is the finest and most difficult prediction target, while family and order represent coarser taxonomic levels.

4.2.2 Taxonomy and Attribute Annotations

Following the H-CAST setting, the CUB-200-2011 species are organized into a three-level hierarchy consisting of 13 orders, 38 families, and 200 species [25]. Each species belongs to exactly one family, and each family belongs to exactly one order. Therefore, each image has a valid taxonomy path:

$$y^o \rightarrow y^f \rightarrow y^s, \tag{4.1}$$

where y^o, y^f, and y^s denote the order, family, and species labels, respectively.

In addition to class labels, CUB-200-2011 provides 312 visual attributes. These attributes describe bird traits such as bill shape, wing color, breast pattern, tail shape, head pattern, eye color, belly color, crown color, and other part-based properties. In this thesis, the 312 attributes are organized into 28 semantic groups based on their attribute names. Each group corresponds to one type of bird trait, and each attribute within a group represents one possible state of that trait.

The proposed model does not use attributes as manual inputs during inference. Instead, class-level attribute annotations are used during training as auxiliary supervision. For an image belonging to species c, the attribute target is obtained from the class-level attribute vector of that species. This setting is consistent with the goal of improving image-based hierarchical classification without requiring users to provide attribute labels at test time.

Several attribute usage strategies are evaluated in this chapter. **FullAttr-312** uses all 312 CUB attributes. **Sibling-IVAS-188** is an internal node-vs-siblings statistical selection baseline implemented during this thesis. It uses train-only image-level binary attribute annotations and ranks attributes primarily by positive occurrence-rate difference between a target taxonomy node and its corresponding negative set, with Information Value used as a secondary criterion. **HSDAS-199** selects attributes based on species-level hard-sibling discrimination using class-level continuous attribute profiles. **Random-199** is used as a control to test whether the benefit comes from the HSDAS selection criterion or merely from reducing the number of attributes.

4.2.3 Train/Test Protocol

The main results in this thesis follow the official CUB-200-2011 train/test protocol. The model is trained on the official training split and evaluated on the official test split. No separate validation set is used for the main reported comparison. This choice follows the experimental setting of H-CAST and allows a direct comparison between the proposed attribute-guided model and the image-only H-CAST baseline.

This protocol is important because the original CUB-200-2011 benchmark and the H-CAST experiments report results using the official train/test split. If a separate validation set were created by splitting the training set, the model would be trained on fewer images, and the resulting numbers would not be directly comparable to the H-CAST baseline trained on the full official training split. Therefore, all main tables in this chapter are reported under the official train/test setting.

4.2.4 Implementation Details

The implementation is based on the original CAST/H-CAST codebase. The H-CAST backbone and the hierarchical visual parsing pipeline are reused and adapted for the CUB-200-2011 setting. The main components implemented in this thesis are the attribute-

group representation, HSDAS attribute selection, attribute-group queries, group-specific attribute prediction heads, species-level attribute fusion, and the attribute attention diversity loss.

Unless otherwise specified, the experiments use the `cast_small` model configuration. The model uses a convolutional stem, a patch size of 8, an embedding dimension of 384, and four hierarchical stages. The numbers of graph clusters are set to $[64, 32, 16, 8]$, and the Transformer block depths are $[3, 3, 3, 2]$. The model predicts three hierarchy levels: species, family, and order.

For attribute-guided experiments, the attribute mask is loaded before model construction. If a selected attribute mask is provided, only the retained attribute states are used to define the active attribute groups and the output dimensions of the group-specific attribute classifiers. If no mask is provided, the model uses all 312 attributes grouped into the original 28 CUB attribute groups.

The training objective consists of hierarchical classification loss, auxiliary attribute prediction loss, optional Tree-Path KL divergence loss, and optional attribute attention diversity loss. The attribute loss weight is denoted by λ_{attr}, and the diversity loss weight is denoted by λ_{div}. The default attribute-guided setting uses $\lambda_{attr} = 0.5$. Different values of λ_{div} are evaluated in the ablation study.

The model is trained using the AdamW optimizer with cosine learning-rate scheduling. Standard data augmentation techniques from the H-CAST/DeiT training pipeline are used, including random augmentation, Mixup, CutMix, label smoothing, and random erasing when enabled by the training configuration. The input image size is 224×224. Evaluation is performed on the official test set using the image and its corresponding superpixel segmentation input required by the CAST/H-CAST backbone.

During training, the model receives images, superpixel segmentations, taxonomy labels, and class-level attribute targets. During inference, the model receives only the image because the superpixel segmentation required by CAST/H-CAST is generated automatically from the image and no ground-truth attribute labels are provided at inference time.

4.3 Compared Methods

This section describes the methods compared in the experiments. All reported results in the main comparison are obtained under the official CUB-200-2011 train/test protocol.

Table 4.1: Image-only baselines used for comparison. Results are reported under the official CUB train/test protocol.

Method	Description
H-CAST reported by authors	Image-only H-CAST result reported in the original paper. Used as an external reference.
H-CAST rerun	Image-only H-CAST model rerun in this thesis under the same CUB train/test protocol. Used as the main image-only baseline.

The compared methods are divided into three groups. The first group contains image-only baselines. The second group contains attribute-guided model variants. The third group compares different attribute selection strategies used to construct the attribute supervision mask.

4.3.1 Image-only Baselines

The image-only baselines do not use CUB attribute annotations during training or inference. They are included to evaluate whether attribute-guided learning provides additional benefit over a standard hierarchical image classifier.

H-CAST reported by the authors. This is the result reported in the original H-CAST paper. It serves as the external reference point for the image-only hierarchical classification setting. Since this result comes from the original paper, it is included only for comparison and is not produced by the implementation in this thesis.

H-CAST rerun. This baseline is obtained by rerunning the image-only H-CAST model under the same CUB train/test protocol used in this thesis. It is the most important image-only baseline because it is trained and evaluated in the same experimental environment as the proposed attribute-guided models. Comparing against this rerun helps separate the effect of the proposed attribute module from differences caused by implementation or training environment.

4.3.2 Attribute-guided Variants

The attribute-guided variants extend the image-only H-CAST model by adding attribute-oriented learning to the species-level branch. In all these variants, attribute labels are

Table 4.2: Attribute-guided variants considered in the experiments.

Variant	λ_{attr}	λ_{div}	**Purpose**
Attribute-guided H-CAST w/o diversity	0.5	0.0	Tests the effect of attribute supervision and species-level fusion without diversity regularization.
Attribute-guided H-CAST with diversity	0.5	0.5	Main proposed setting with attribute supervision, species-level fusion, and diversity loss.
Attribute-guided H-CAST with weak diversity	0.5	0.1	Tests whether a smaller diversity weight gives a better trade-off.
Attribute-guided H-CAST with stronger diversity	0.5	1.0	Tests whether stronger diversity regularization improves or hurts performance.

used only during training. During inference, the model receives only the input image and the superpixel segmentation required by the H-CAST/CAST backbone.

Attribute-guided H-CAST. This is the main model proposed in this thesis. It uses one learnable query for each active attribute group. These queries attend to species-level graph-node features and produce attribute-oriented representations. The representations are supervised by group-specific attribute classifiers and are also fused with the species-level class token before species prediction.

Attribute-guided H-CAST without diversity loss. This variant removes the attribute attention diversity loss by setting $\lambda_{div} = 0$. It is used to evaluate whether the diversity regularization term contributes to the final performance.

Attribute-guided H-CAST with diversity loss. This variant enables the diversity term with $\lambda_{div} > 0$. It is used to evaluate whether encouraging different attribute macro-parts to attend to complementary graph nodes improves fine-grained classification.

4.3.3 Attribute Selection Variants

This thesis evaluates several ways of choosing the attribute subset used by the attribute-guided module. These variants are important because CUB contains 312 attributes, but not all of them are equally useful for species-level discrimination.

FullAttr-312. This setting uses all 312 CUB attributes. It preserves the full semantic

information available in the dataset, but may include redundant or weakly discriminative attributes.

HSDAS-199. This is the main attribute selection strategy studied in this thesis. HSDAS selects attributes at the species level by combining general sibling discriminativeness and hard-sibling discriminativeness. It uses class-level continuous attribute profiles and prioritizes attributes that distinguish a species from sibling species with similar attribute profiles.

HSDAS-174. This variant uses a smaller HSDAS-selected subset. It is included to analyze how attribute subset size affects classification performance.

Sibling-IVAS-188. This is an internal statistical attribute-selection baseline implemented during this thesis. It uses train-only image-level binary attribute annotations and compares each taxonomy node with a corresponding negative set. For species-level nodes, the negative set consists of sibling species within the same family. Attributes are ranked primarily by positive occurrence-rate difference between the target node and the negative set, with Information Value used as a secondary criterion. This baseline is included to compare HSDAS against another taxonomy-aware but non-hard-sibling selection strategy.

Random-199. This setting selects 199 random attributes, matching the number of attributes in HSDAS-199. It is used as a control to test whether performance changes come from the HSDAS selection criterion or merely from reducing the number of attributes.

4.4 Evaluation Metrics

The models are evaluated using both classification accuracy and hierarchy-consistency metrics, following the H-CAST evaluation protocol [25]. Since the task is hierarchical, species accuracy alone is not sufficient. A model may predict the correct species, family, or order independently, but the predicted labels may not form a valid taxonomy path. Therefore, this thesis reports level-wise accuracy, Full-Path Accuracy, Weighted Average Precision, Tree-based InConsistency Error, and attribute prediction accuracy.

4.4.1 Level-wise Accuracy

Level-wise accuracy measures the classification accuracy at each taxonomy level independently. In this thesis, the three reported level-wise accuracies are species accuracy, family

Table 4.3: Attribute selection variants compared in this thesis. All variants are evaluated under the official CUB train/test protocol.

Method	**#Attr**	**Description**
FullAttr-312	312	Uses all CUB attributes.
HSDAS-199	199	Main hard-sibling-aware selected subset using class-level continuous attributes.
HSDAS-174	174	Smaller HSDAS-selected subset for subset-size analysis.
Sibling-IVAS-188	188	Internal train-only node-vs-siblings statistical baseline using positive occurrence-rate difference and Information Value.
Random-199	199	Random subset control with the same size as HSDAS-199.

accuracy, and order accuracy.

Let N be the number of test samples. For each sample i, let y_i^s, y_i^f, and y_i^o be the ground-truth species, family, and order labels. Let $\hat{y}_i^s$, $\hat{y}_i^f$, and $\hat{y}_i^o$ be the corresponding predictions. The species accuracy is defined as

$$Acc_s = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left[\hat{y}_i^s = y_i^s\right], \tag{4.2}$$

where $\mathbb{I}[\cdot]$ is the indicator function. Similarly, family accuracy and order accuracy are defined as

$$Acc_f = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left[\hat{y}_i^f = y_i^f\right], \tag{4.3}$$

and

$$Acc_o = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left[\hat{y}_i^o = y_i^o\right]. \tag{4.4}$$

Species accuracy is the main fine-grained classification metric because species is the most specific label. Family and order accuracies measure performance at coarser levels. In the proposed model, the attribute-guided fusion is applied to the species-level branch, so the largest improvement is expected at the species level.

4.4.2 Weighted Average Precision

In addition to level-wise accuracy, this thesis reports weighted Average Precision (wAP), following the H-CAST evaluation protocol. Although named weighted Average Precision following H-CAST, this metric is computed as a class-count-weighted average of level-wise Top-1 accuracies. Let L be the number of hierarchy levels. For each level l, let N_l denote the number of classes at that level and let P_l denote the Top-1 accuracy at that level. The weighted Average Precision is defined as

$$wAP = \sum_{l=1}^{L} \frac{N_l}{\sum_{k=1}^{L} N_k} P_l. \tag{4.5}$$

This metric weights each level according to the number of classes at that level. For CUB-200-2011, the hierarchy contains 13 orders, 38 families, and 200 species. Therefore, wAP is more strongly influenced by species accuracy than by order or family accuracy. This makes wAP useful as a complementary metric for evaluating hierarchical classification, especially when the finest level contains many more classes than the coarser levels.

However, wAP does not directly measure whether the predicted labels form a valid taxonomy path. Therefore, it should be interpreted together with FPA and TICE. In this thesis, FPA remains the main hierarchical correctness metric, while wAP provides an additional weighted summary of level-wise accuracy.

4.4.3 Full-Path Accuracy

Full-Path Accuracy (FPA) measures whether the model predicts the entire taxonomy path correctly. A prediction is counted as correct only when the species, family, and order labels are all correct.

Formally, FPA is defined as

$$FPA = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left[\hat{y}_i^s = y_i^s \wedge \hat{y}_i^f = y_i^f \wedge \hat{y}_i^o = y_i^o\right]. \tag{4.6}$$

FPA is stricter than level-wise accuracy because a sample is counted as correct only if all levels are correct simultaneously. This metric is important for hierarchical classification because it evaluates whether the model can produce a fully correct taxonomic prediction. Since family and order accuracies are usually high on CUB, changes in FPA are often strongly influenced by changes in species accuracy.

4.4.4 Tree-based InConsistency Error

Tree-based InConsistency Error (TICE) measures whether the predicted labels form a valid path in the hierarchy. Unlike FPA, TICE does not require the prediction to match the ground truth. Instead, it checks whether the predicted species, family, and order are mutually consistent according to the taxonomy.

Let $\pi_f(\cdot)$ denote the parent-family mapping from species to family, and let $\pi_o(\cdot)$ denote the parent-order mapping from family to order. A predicted path is consistent if

$$\hat{y}_i^f = \pi_f(\hat{y}_i^s) \quad \text{and} \quad \hat{y}_i^o = \pi_o(\hat{y}_i^f). \tag{4.7}$$

The inconsistency indicator for sample i is defined as

$$e_i = \mathbb{I}\left[\hat{y}_i^f \neq \pi_f(\hat{y}_i^s) \ \vee \ \hat{y}_i^o \neq \pi_o(\hat{y}_i^f)\right]. \tag{4.8}$$

TICE is then computed as

$$TICE = \frac{1}{N}\sum_{i=1}^{N} e_i. \tag{4.9}$$

A lower TICE indicates better taxonomy consistency. A model can have high level-wise accuracy but still produce inconsistent paths if its species, family, and order heads make predictions that do not correspond to the same taxonomy branch. Therefore, TICE complements accuracy-based metrics by measuring the structural validity of the prediction path.

In the proposed model, TICE is especially useful because the attribute-guided fusion is applied mainly to the species branch. If the species prediction changes without a corresponding change in the family or order predictions, taxonomy inconsistency may increase. Thus, TICE helps analyze the trade-off between improving fine-grained species discrimination and preserving hierarchical consistency.

4.4.5 Attribute Prediction Accuracy

Attribute prediction accuracy evaluates the auxiliary attribute prediction heads. It is reported only for attribute-guided models. Since the model uses group-specific classifiers, attribute accuracy is computed over active attribute groups rather than over a single 312-dimensional binary vector.

For each active group g, the model predicts one retained attribute state. Let $\hat{t}_{i,g}$ be the predicted attribute state for sample i and group g, and let $\tilde{t}_{i,g}$ be the remapped ground-truth attribute target. If the target state is not retained in the selected attribute mask, the sample-group pair is ignored.

For each mini-batch, attribute accuracy is first computed separately for each active group with at least one valid target. Let Ω_g denote the set of valid samples for group g in the mini-batch. The group accuracy is

$$Acc_{attr}^{g} = \frac{1}{|\Omega_g|} \sum_{i \in \Omega_g} \mathbb{I}\left[\hat{t}_{i,g} = \tilde{t}_{i,g}\right]. \tag{4.10}$$

The batch attribute accuracy is then computed as the average over valid groups:

$$Acc_{attr}^{batch} = \frac{1}{|\mathcal{G}_{valid}|} \sum_{g \in \mathcal{G}_{valid}} Acc_{attr}^{g}. \tag{4.11}$$

The reported attribute accuracy is the average of this batch-level attribute accuracy over the test set, weighted by batch size in the evaluation logger.

This metric measures how well the auxiliary attribute heads learn the selected attribute targets. However, attribute prediction accuracy should not be interpreted as the final task performance. A higher attribute accuracy does not always guarantee higher species accuracy, because some attributes may be easy to predict but not highly discriminative for fine-grained species classification. Therefore, attribute accuracy is used as a diagnostic metric to analyze the behavior of the attribute branch, while species accuracy, FPA, wAP, and TICE remain the primary metrics for evaluating hierarchical classification.

4.5 Main Results

This section presents the main comparison between the proposed attribute-guided H-CAST model and image-only baselines under the official CUB-200-2011 train/test protocol. The goal is to answer RQ1: whether attribute-guided learning improves hierarchical bird classification compared with the image-only H-CAST baseline. Unless otherwise stated, the proposed attribute-guided H-CAST refers to the HSDAS-199 setting with $\lambda_{attr} = 0.5$ and $\lambda_{div} = 0.5$.

Table 4.4 reports Full-Path Accuracy (FPA), level-wise accuracies, Weighted Average Precision (wAP), Tree-based InConsistency Error (TICE), and attribute prediction accuracy

Table 4.4: Main comparison under the official CUB-200-2011 train/test protocol. Attribute accuracy is reported only for attribute-guided models.

Method	**FPA**	**Species**	**Family**	**Order**	**wAP**	**TICE**	**Attr**
H-CAST authors	83.28	84.86	95.12	98.65	87.13	4.12	–
H-CAST rerun	82.86	84.55	95.22	98.74	86.90	4.35	–
Proposed attribute-guided H-CAST	**84.36**	**85.83**	**95.62**	**99.03**	**88.00**	**3.62**	87.04

when applicable. The image-only baselines do not use attribute annotations and therefore do not report attribute accuracy.

The proposed attribute-guided H-CAST improves all accuracy-based metrics compared with the H-CAST rerun. Specifically, it improves FPA from 82.86% to 84.36%, species accuracy from 84.55% to 85.83%, family accuracy from 95.22% to 95.62%, order accuracy from 98.74% to 99.03%, and wAP from 86.90% to 88.00%. It also reduces TICE from 4.35% to 3.62%. This result suggests that adding attribute-guided auxiliary supervision and species-level attribute fusion provides useful information beyond the image-only H-CAST baseline.

The largest improvement appears at the species level, which is consistent with the design of the proposed architecture. Attribute-guided fusion is applied to the species-level branch, where fine-grained cues are most important. Since wAP is weighted by the number of classes at each hierarchy level, the improvement in species accuracy also contributes strongly to the improvement in wAP.

Compared with the result reported by the original H-CAST authors, the proposed model also achieves higher FPA, species accuracy, family accuracy, order accuracy, and wAP, while obtaining a lower TICE. However, the comparison with the H-CAST rerun should be considered the primary comparison, because it is produced under the same implementation environment and training setup as the proposed model.

Overall, the main result supports the effectiveness of attribute-guided learning for hierarchical bird classification. Nevertheless, later ablation studies show that the improvement depends on the attribute subset and the loss configuration. Therefore, the proposed method should be interpreted as improving fine-grained species discrimination, while the relationship between species accuracy and taxonomy consistency still requires careful anal-

Table 4.5: Effect of attribute selection strategy under the official CUB train/test protocol. Random-199 is reported as the mean over three random seeds.

Method	#Attr	FPA	Species	Family	Order	wAP	TICE	Attr
FullAttr-312	312	84.29	85.59	95.65	98.81	87.80	**3.25**	85.65
Sibling-IVAS-188	188	84.02	85.28	95.69	98.81	87.56	3.52	86.95
Random-199 mean	199	84.15	85.45	**95.79**	98.95	87.71	3.33	**91.28**
HSDAS-199	199	**84.36**	**85.83**	95.62	**99.03**	**88.00**	3.62	87.04

ysis.

4.6 Ablation Studies

This section analyzes the contribution of different components of the proposed method. All ablation studies follow the official CUB-200-2011 train/test protocol. The goal is to understand how attribute selection, attribute subset size, diversity loss, and random attribute selection affect the final performance. Unless otherwise stated, attribute-guided variants use species-level fusion with $\lambda_{attr} = 0.5$ and $\lambda_{div} = 0.5$. The diversity-loss ablation varies only λ_{div}, while the fusion-design ablation varies the use of fusion, attribute supervision, and diversity regularization.

4.6.1 Effect of Attribute Selection

This ablation evaluates different attribute usage strategies. It addresses RQ2 and RQ3 by comparing compact selected subsets with FullAttr-312 and by comparing HSDAS with Random-199 and Sibling-IVAS-188.

Among the evaluated attribute usage strategies shown in Table 4.5, HSDAS-199 achieves the highest species accuracy and wAP. It also achieves higher FPA than FullAttr-312, Sibling-IVAS-188, and the Random-199 mean, although Random-199 seed 2 slightly exceeds it in FPA as discussed later. Compared with FullAttr-312, HSDAS-199 improves species accuracy from 85.59% to 85.83% and wAP from 87.80% to 88.00%, while using fewer attributes. This suggests that a compact hard-sibling-aware subset can be more useful for species-level discrimination than using all available attributes. However, the improvement over FullAttr-312 is moderate, and FullAttr-312 obtains a lower TICE. Therefore, HSDAS-199 should not be interpreted as strictly better in all aspects.

Compared with Sibling-IVAS-188, HSDAS-199 improves FPA, species accuracy, order ac-

curacy, and wAP. Sibling-IVAS-188 is already a taxonomy-aware internal baseline because it compares each node with sibling-based negative sets. Therefore, the improvement of HSDAS over Sibling-IVAS suggests that explicitly emphasizing hard siblings with similar class-level attribute profiles provides additional benefit for fine-grained species discrimination.

The comparison with Random-199 should be interpreted carefully. HSDAS-199 improves over the Random-199 average in FPA, species accuracy, order accuracy, and wAP. It also achieves higher species accuracy than all three random seeds. However, Random-199 obtains better family accuracy and lower TICE on average, and one random seed obtains slightly higher FPA than HSDAS-199. Therefore, the result supports HSDAS as a useful species-discrimination-oriented selection strategy, but it does not support a strong claim that HSDAS dominates random selection in every metric.

Attribute prediction accuracy is reported only as a diagnostic metric. Random-199 obtains the highest attribute prediction accuracy, but this does not translate into the best species accuracy or wAP. This shows that attribute prediction accuracy and final classification performance are not perfectly correlated. A random subset may contain attribute states that are easier to predict, while HSDAS selects attributes according to hard-sibling discrimination.

4.6.2 Effect of Attribute Subset Size

This ablation studies how the number of selected attributes affects performance. In this experiment, the HSDAS selection strategy is evaluated with different top-k settings, with the results shown in Table 4.6. **HSDAS-174** is obtained by selecting the top 3 attributes for each species and then taking the union of selected attributes. **HSDAS-199** is obtained by selecting the top 5 attributes for each species. FullAttr-312 is included as the reference setting that uses all CUB attributes.

The results show that increasing HSDAS selection from top-3 to top-5 attributes per species gives a small improvement in FPA, species accuracy, order accuracy, and wAP. HSDAS top-5 improves species accuracy from 85.78% to 85.83% and wAP from 87.93% to 88.00%. The gain is modest, which suggests that HSDAS top-3 already preserves most of the useful attribute supervision.

However, the improvement is not uniform across all metrics. HSDAS top-3 obtains lower

Table 4.6: Effect of HSDAS attribute subset size. HSDAS-174 uses top-3 attributes per species, while HSDAS-199 uses top-5 attributes per species. All settings use the official CUB train/test protocol.

Method	#Attr	FPA	Species	Family	Order	wAP	TICE	Attr
HSDAS top-3 (174)	174	84.29	85.78	95.50	98.88	87.93	3.37	**87.42**
HSDAS top-5 (199)	199	**84.36**	**85.83**	95.62	**99.03**	**88.00**	3.62	87.04
FullAttr-312	312	84.29	85.59	**95.65**	98.81	87.80	**3.25**	85.65

Table 4.7: Effect of the attribute attention diversity loss using HSDAS-199.

Setting	λ_{div}	FPA	Species	Family	Order	wAP	TICE	Attr
HSDAS-199 without diversity	0.0	84.16	85.47	95.63	98.62	87.69	**3.14**	87.01
HSDAS-199 weak diversity	0.1	84.02	85.33	**95.79**	98.91	87.62	3.44	**87.46**
HSDAS-199 main setting	0.5	**84.36**	**85.83**	95.62	**99.03**	**88.00**	3.62	87.04
HSDAS-199 strong diversity	1.0	84.04	85.33	95.75	98.79	87.60	3.40	86.12

TICE and higher attribute prediction accuracy than HSDAS top-5. This suggests that the more compact top-3 subset may be easier for the attribute branch to predict and is associated with lower TICE in this experiment. In contrast, the top-5 subset slightly improves species-level discrimination and wAP but increases TICE.

Compared with FullAttr-312, both HSDAS subsets achieve higher species accuracy and wAP while using fewer attributes. FullAttr-312 provides the best TICE and slightly higher family accuracy, but its species accuracy and wAP are lower than both HSDAS variants. Overall, HSDAS top-5 is selected as the main proposed setting because it achieves the best species accuracy, FPA, and wAP among the subset-size variants. However, HSDAS top-3 remains a competitive compact alternative, with better attribute prediction accuracy and lower TICE.

4.6.3 Effect of Attribute Diversity Loss

This ablation evaluates the role of the attribute attention diversity loss. It contributes to RQ4 by analyzing how diversity regularization affects species accuracy, FPA, wAP, and taxonomy consistency.

The results in Table 4.7 suggest that the diversity loss has a non-monotonic effect. The main setting with $\lambda_{div} = 0.5$ achieves the best FPA and species accuracy, suggesting that a moderate amount of diversity regularization can help fine-grained species discrimination.

However, weaker diversity ($\lambda_{div} = 0.1$) and stronger diversity ($\lambda_{div} = 1.0$) do not improve species accuracy over the no-diversity setting. Therefore, the results do not support the claim that simply increasing diversity regularization always improves performance.

The no-diversity setting obtains the lowest TICE. This indicates that the model can produce more taxonomically consistent paths when the diversity regularizer is removed. In contrast, the best species accuracy is obtained with $\lambda_{div} = 0.5$, which has higher TICE. This suggests a trade-off between species-level discrimination and taxonomy consistency.

One possible explanation is that the diversity loss is intended to encourage attribute macro-parts to use more complementary graph-node evidence, which may improve species prediction. However, because family and order predictions are not directly fused with attribute features, changes in species prediction may not always be matched by corresponding changes in the coarser predictions. This can increase path inconsistency even when species accuracy improves.

The result with $\lambda_{div} = 1.0$ suggests that too much diversity regularization may become harmful. A strong diversity constraint may force attention maps to spread or separate even when the image contains only a few truly discriminative cues. Therefore, the diversity loss should be treated as a regularizer that requires careful weighting rather than a universally beneficial component.

4.6.4 Effect of Attribute Fusion Design

This ablation directly addresses RQ4 by analyzing the contribution of auxiliary attribute supervision, species-level attribute fusion, and diversity regularization. The results are shown in Table 4.8.

The no-fusion setting shows that auxiliary attribute supervision alone is not sufficient to improve species classification. Although this setting obtains the highest attribute prediction accuracy, its species accuracy is 84.38%, which is slightly lower than the H-CAST rerun. This indicates that learning to predict attributes does not automatically transfer to better species recognition unless attribute-oriented features are explicitly connected to the species classifier.

The comparison between the no-fusion setting and the fusion setting with $\lambda_{attr} = 0.5$, $\lambda_{div} = 0$ shows the importance of species-level fusion. With the same attribute supervision weight and no diversity loss, adding fusion improves FPA from 83.09% to 84.16%, species

Table 4.8: Effect of attribute supervision, species-level fusion, and diversity regularization under the official CUB train/test protocol.

Method	**FPA**	**Species**	**Family**	**Order**	**wAP**	**TICE**	**Attr**
H-CAST rerun	82.86	84.55	95.22	98.74	86.90	4.35	–
No fusion $\lambda_{attr} = 0.5, \lambda_{div} = 0$	83.09	84.38	95.58	98.72	86.82	3.54	**89.04**
Fusion $\lambda_{attr} = 0, \lambda_{div} = 0$	83.73	85.21	95.53	98.74	87.47	3.87	23.02
Fusion $\lambda_{attr} = 0.5, \lambda_{div} = 0$	84.16	85.47	**95.63**	98.62	87.69	**3.14**	87.01
Fusion $\lambda_{attr} = 0.5, \lambda_{div} = 0.5$	**84.36**	**85.83**	95.62	**99.03**	**88.00**	3.62	87.04

accuracy from 84.38% to 85.47%, and wAP from 86.82% to 87.69%. This supports the design choice of concatenating the attention-pooled attribute summary with the species class token.

The fusion-only setting, where $\lambda_{attr} = 0$, improves species accuracy compared with the no-fusion setting, but its attribute accuracy drops sharply to 23.02%. This suggests that the additional fusion path can learn classification-useful features through the species classification loss, but these features are not semantically aligned with CUB attributes when attribute supervision is removed.

When attribute supervision is added to the fusion model with $\lambda_{attr} = 0.5$, FPA increases from 83.73% to 84.16%, species accuracy increases from 85.21% to 85.47%, TICE decreases from 3.87% to 3.14%, and attribute accuracy increases from 23.02% to 87.01%. This suggests that attribute supervision helps align the fused representation with semantic bird traits and improves taxonomy consistency in the no-diversity setting.

Adding diversity regularization with $\lambda_{div} = 0.5$ further improves FPA, species accuracy, order accuracy, and wAP. The full model achieves the best species accuracy and wAP among the fusion-design settings. However, TICE increases compared with the no-diversity fusion setting, which again indicates a trade-off between fine-grained species discrimination and taxonomy consistency.

Overall, this ablation shows that the proposed model benefits from the combination of attribute supervision and species-level fusion. Attribute supervision alone learns meaningful attribute predictors but does not improve species recognition without fusion. Fusion allows attribute-oriented features to affect the species classifier, while attribute supervi-

Table 4.9: Random attribute control. Random-199 uses the same number of attributes as HSDAS-199 and preserves the same number of selected attribute states per group. Mean and standard deviation are computed over three random seeds.

Method	**#Attr**	**FPA**	**Species**	**Family**	**Order**	**wAP**	**TICE**	**Attr**
Random-199 seed 1	199	83.85	85.21	95.75	98.96	87.52	3.47	91.02
Random-199 seed 2	199	84.40	85.66	95.86	98.95	87.89	3.42	91.06
Random-199 seed 3	199	84.19	85.47	95.75	98.95	87.72	3.11	91.77
Random-199 mean ± std	199	84.15 ±0.28	85.45 ±0.23	95.79 ±0.06	98.95 ±0.01	87.71 ±0.19	3.33 ±0.20	91.28 ±0.42
HSDAS-199	199	**84.36**	**85.83**	95.62	**99.03**	**88.00**	3.62	87.04

sion encourages these features to align with the selected CUB attribute targets. The best species accuracy and wAP are obtained when attribute supervision, fusion, and diversity regularization are used together.

4.6.5 Random Attribute Control

Random-199 is used to test whether the performance of HSDAS-199 comes from the hard-sibling-aware selection criterion or simply from using fewer attributes than FullAttr-312. In this thesis, Random-199 is constructed as a group-matched random control. It does not sample 199 attributes uniformly from the full 312-dimensional attribute space. Instead, for each semantic attribute group, it randomly samples the same number of attribute states as the HSDAS-199 mask. Therefore, Random-199 preserves the same group-level structure and active-group coverage as HSDAS-199, while removing the hard-sibling-aware selection criterion.

The Random-199 control in Table 4.9 shows that group-matched random subsets can be competitive. This is expected because Random-199 keeps the same number of attribute states in each semantic group as HSDAS-199. As a result, it still preserves broad coverage over color, shape, pattern, and part-related attribute groups. Therefore, this control is stronger than a purely uniform random subset over all 312 attributes, where the group distribution could become unbalanced.

Random-199 seed 2 obtains slightly higher FPA than HSDAS-199. However, this does not contradict the main conclusion of this thesis. FPA depends on the joint correctness of species, family, and order predictions. HSDAS-199 achieves higher species accuracy than seed 2, improving from 85.66% to 85.83%, and also achieves higher wAP, improving from

87.89% to 88.00%. These metrics are more directly related to fine-grained species discrimination, which is the main goal of the proposed attribute selection strategy. In contrast, seed 2 obtains higher family accuracy and lower TICE, which can lead to slightly better full-path correctness in this particular run. The FPA difference is only 0.04 percentage points, corresponding to roughly two test images. Therefore, this result is better interpreted as random variation among strong group-matched attribute subsets rather than evidence that random selection is superior.

Across the three random seeds, HSDAS-199 achieves higher species accuracy and wAP than every Random-199 run. It also achieves higher FPA than the Random-199 mean. This suggests that hard-sibling-aware selection provides a useful bias toward attributes that are helpful for species-level recognition. At the same time, the advantage is moderate rather than large. Since Random-199 keeps 199 out of 312 attributes and preserves the same group-level coverage as HSDAS-199, it can still contain many useful attribute states.

The TICE results also show that HSDAS-199 should not be claimed to dominate random selection in every metric. All three Random-199 seeds obtain lower TICE than HSDAS-199. This means that the random subsets can produce more taxonomically consistent prediction paths in some runs, even though they do not achieve the best species accuracy or wAP. This observation is consistent with the broader trend in the ablation studies: improving species-level discrimination does not always lead to the lowest taxonomy inconsistency.

The attribute accuracy of Random-199 is also higher than that of HSDAS-199. This result should be interpreted carefully. Attribute prediction accuracy measures how well the auxiliary heads predict the selected attribute states, but it does not directly measure whether those attributes are useful for species classification. Random masks may include more frequent or easier-to-predict attribute states, allowing the auxiliary attribute heads to achieve higher accuracy. In contrast, HSDAS selects attributes according to hard-sibling discriminativeness, which may produce harder attribute targets but more species-relevant supervision. This explains why HSDAS-199 has lower attribute prediction accuracy than Random-199 but higher species accuracy and wAP.

Overall, the random control shows that attribute subset size and group-level coverage already provide a strong baseline. HSDAS-199 should therefore not be interpreted as outperforming random selection in every metric. Its advantage is more specific: it provides a targeted and interpretable hard-sibling-aware selection strategy that achieves

higher species accuracy and wAP than all three group-matched random subsets, while also achieving higher FPA than the Random-199 mean.

4.7 Qualitative and Diagnostic Analysis

4.7.1 Prediction-Level Case Analysis

To further analyze the species-level improvement, we compare the prediction outputs of the image-only H-CAST rerun and the proposed attribute-guided model on the CUB test set.

Table 4.10 summarizes the prediction-level case analysis. The proposed model corrects 236 species-level errors made by the H-CAST rerun. Among these corrected cases, 194 correspond to intra-family confusions, where H-CAST predicts the correct family but the wrong species. This accounts for approximately 82.2% of the corrected species cases. This observation supports the motivation of the proposed hard-sibling-aware attribute selection strategy, because many corrected errors occur between visually related species within the same family.

However, the improvement is not one-sided. The proposed model also introduces 161 new species-level errors where H-CAST was originally correct. This means that the attribute-guided branch changes the species-level decision boundary rather than simply adding correct predictions on top of the baseline. Nevertheless, the number of corrected errors is larger than the number of newly introduced errors, resulting in a net gain of 75 correctly classified species. Over 5,794 test images, this corresponds to approximately 1.29 percentage points, which is consistent with the observed species accuracy improvement from 84.55% to 85.83%.

At the same time, there remain 659 images where both models predict the wrong species. This indicates that the proposed attribute-guided module does not eliminate all fine-grained recognition errors. In addition, 86 images are cases where the proposed model predicts the correct species but the full hierarchical path is still inconsistent. This helps explain why the proposed model improves species accuracy and wAP, while its TICE is not the lowest among all ablation settings.

Figure 4.1 shows representative intra-family confusion cases corrected by the proposed model. These examples illustrate the type of errors captured by the second row of Table 4.10: the image-only H-CAST baseline already predicts the correct family but fails to

Table 4.10: Prediction-level case analysis comparing the H-CAST rerun and the proposed attribute-guided model on the CUB test set.

Case type	**Count**	**Interpretation**
H-CAST wrong, proposed correct	236	Species-level errors corrected by the proposed model.
H-CAST wrong within same family, proposed correct	194	Corrected intra-family species confusions.
H-CAST correct, proposed wrong	161	Correct baseline predictions lost by the proposed model.
Both models wrong at species level	659	Remaining hard failure cases.
Proposed species correct but path inconsistent	86	Species prediction is correct, but the predicted hierarchical path is not fully consistent.
Net species-level gain	+75	Corrected errors minus newly introduced errors.

distinguish the correct species, while the proposed model makes the correct species-level prediction. This supports the quantitative observation that the proposed attribute-guided model is particularly useful for some fine-grained species confusions within the same family.

It should be noted that these examples are illustrative rather than exhaustive. The main evidence for the improvement comes from the prediction-level counts and the quantitative metrics reported in the previous sections.

4.7.2 Diagnostic Analysis of Attribute Attention

In addition to the prediction-level case analysis, we inspect the internal attention behavior of the proposed attribute-guided module. This analysis is intended to provide diagnostic insight into how the attribute branch behaves, rather than to prove that the model accurately localizes semantic attributes.

The proposed module produces two attention signals. First, during species-level fusion, the species class token attends to the attribute-group features through an attention-pooling operation. This produces a distribution over active attribute groups and indicates which groups contribute more strongly to the fused species representation. Second, each



Figure 4.1: Representative intra-family species confusions corrected by the proposed model. In each example, H-CAST predicts an incorrect species while still predicting the correct family, whereas the proposed attribute-guided model predicts the correct species.

attribute-group query cross-attends to the species-level graph nodes generated by the H-CAST backbone. These graph-node attention weights are back-projected to the initial superpixels using the H-CAST graph assignment structure and are overlaid on the input image for visualization.

For the three-level CUB hierarchy used in this thesis, the attribute-group queries attend to the species-level graph representation. The visualized images correspond to the evaluation input after resizing and center cropping, which is the same input view used by the model during inference.

The attention ranking analysis shows that the species-token pooling attention is often dominated by global morphology groups, especially *size* and *shape*. In many inspected examples, these two groups receive substantially higher pooling weights than most other groups. This suggests that the fusion module assigns high importance to global morphology-oriented attribute features. However, these weights should not be interpreted as evidence that the model precisely localizes size or shape in the image. Instead, they indicate that the fused species representation may rely on global cues such as morphology, silhouette, scale, or contextual information.

This behavior is partly related to the selected HSDAS-199 mask. In this mask, the *size* group contains only one retained attribute state, while the *shape* group contains only two retained states. Therefore, these groups provide limited within-group attribute classifi-

cation supervision, especially for the singleton *size* group. Their queries can still receive gradients through the species-level fusion path, so high CLS pooling weights may reflect classification-useful latent features rather than precisely localized semantic attribute evidence.

Figure 4.2 shows a representative diagnostic visualization. The CLS pooling chart shows the species-token attention over attribute groups. In this example, the highest-weighted groups are *size*, *shape*, and *back pattern*. Since the top-ranked groups are often dominated by global morphology groups, the spatial heatmaps in Figure 4.2(b) are selected from local or part-related groups for diagnostic inspection. These maps allow us to examine whether some attribute queries attend to plausible bird regions. In some examples, local attribute-query maps overlap with the bird body or discriminative parts. However, this behavior is not sufficiently consistent to support a strong claim of semantic attribute localization.

The spatial attention maps of high-weight global groups further reveal a limitation of the current method. Although *size* and *shape* often receive high pooling weights, their graph-node attention maps are not consistently aligned with a clearly localized semantic part. In some inspected examples, these maps overlap with the bird body or head region, while in other cases they become diffuse or include substantial background regions. Therefore, high pooling weight for global morphology groups should not be interpreted as reliable evidence of precise attribute localization.

Figure 4.3 shows one representative limitation case. In this example, the *size* and *shape* attention maps place a substantial portion of their response on background regions instead of being tightly concentrated on the bird itself. This illustrates that even when these groups receive high CLS pooling weights, the corresponding spatial maps may still reflect a mixture of object cues, silhouette, scale, and contextual information rather than a precisely localized semantic attribute.

For this reason, the attention visualizations are treated as diagnostic evidence rather than localization evidence. The model is trained with class-level attribute supervision and does not use pixel-level attribute localization labels. Therefore, it is not explicitly forced to align each attribute query with the corresponding semantic bird part. The main conclusion of this thesis is based on the quantitative results, ablation studies, and prediction-level case analysis, while the attention visualization is used only to better understand the internal behavior and limitations of the proposed module.

4.7.3 Discussion

The qualitative and diagnostic analyses provide a more detailed view of the quantitative results. The prediction-level case analysis shows that the proposed model corrects more species-level errors than it introduces. Specifically, it corrects 236 species-level errors made by the H-CAST rerun, while introducing 161 new species-level errors where H-CAST was originally correct. This results in a net gain of 75 correctly classified test images. Over 5,794 test images, this corresponds to approximately 1.29 percentage points, which is consistent with the observed species accuracy improvement from 84.55% to 85.83%.

The analysis also shows that many of the corrected cases are intra-family confusions. Among the 236 corrected species-level errors, 194 cases correspond to images where H-CAST predicts the correct family but the wrong species. This accounts for approximately 82.2% of the corrected cases. This observation is consistent with the motivation of the proposed hard-sibling-aware attribute selection strategy, which is designed to prioritize attributes that help distinguish visually related species within the same family.

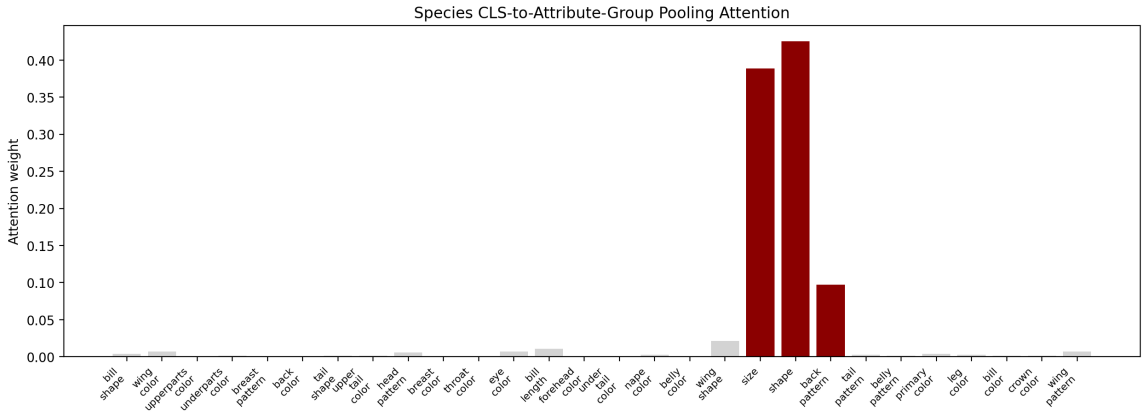
However, the proposed method is not uniformly better than the image-only H-CAST baseline. Some predictions that are correct in H-CAST become incorrect in the proposed model, and 659 images remain misclassified by both models at the species level. This indicates that attribute-guided fusion changes the species-level decision boundary and can improve average performance without improving every individual prediction. Therefore, the proposed method should be interpreted as improving species-level recognition overall, especially for some intra-family confusions, rather than as a model that consistently corrects all fine-grained errors.

The path-consistency analysis also helps explain the trade-off observed in the quantitative results. There are 86 images where the proposed model predicts the correct species but the full hierarchical path is still inconsistent. This is consistent with the fact that the proposed model achieves strong species accuracy, FPA, and wAP in the main setting, but does not obtain the lowest TICE among all ablation settings. Since attribute-guided fusion is applied mainly to the species-level branch, it can improve fine-grained species discrimination without necessarily improving taxonomy consistency in every case.

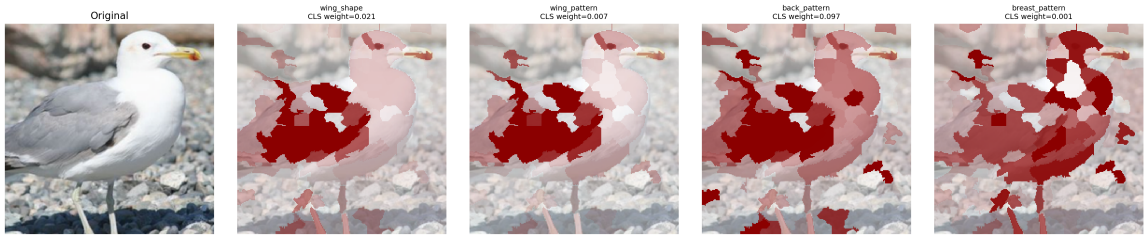
Finally, the attention analysis highlights an important limitation of the current approach. Although the attribute-guided branch improves classification performance, the attention visualizations do not provide strong evidence of precise semantic attribute localization.

Global groups such as *size* and *shape* often dominate the pooling attention, and their spatial maps can be diffuse or include background regions. Some local attribute queries attend to plausible bird regions, but this behavior is not sufficiently consistent to support a strong localization claim. This suggests that the model may use a combination of object-part cues, global morphology, silhouette, scale, and contextual information.

Overall, the qualitative analysis supports the main conclusion of this thesis: attribute-guided fusion improves the main quantitative performance of H-CAST, especially species accuracy, FPA, and wAP, and many of the corrected errors are intra-family species confusions. At the same time, the diagnostic attention analysis shows that the current method should not be interpreted as a precise attribute localization model. Future work could address this limitation by incorporating object masks, part annotations, stronger localization constraints, or attention regularization methods that encourage attribute queries to focus more consistently on bird regions.



(a) Species CLS-to-attribute-group pooling attention.



(b) Selected attribute-query spatial attention maps.

Figure 4.2: Diagnostic visualization of the attribute-guided module. (a) The CLS pooling chart shows the species-token attention over attribute groups. In this example, the three highest-weighted groups are *size*, *shape*, and *back pattern*, indicating that the fusion module assigns strong weight to global morphology-related groups. (b) The spatial heatmaps show selected attribute-query attention maps back-projected to image superpixels. From left to right, the maps correspond to the original image, *wing shape*, *wing pattern*, *back pattern*, and *breast pattern*. These local or part-related groups are selected for diagnostic inspection and are not necessarily the highest-ranked groups in the CLS pooling distribution. These visualizations should not be interpreted as ground-truth attribute localization.

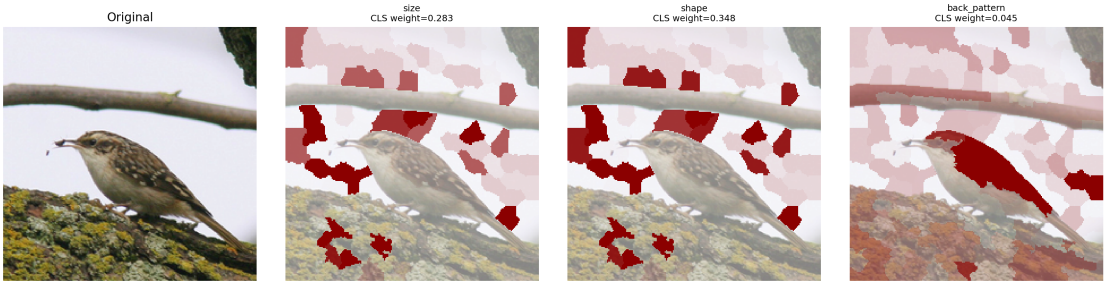


Figure 4.3: Representative limitation case for high-weight global morphology groups. From left to right, the figure shows the original image, the spatial attention map of *shape*, the spatial attention map of *size*, and the spatial attention map of *back pattern*. In this example, the *shape* and *size* queries place substantial attention on background regions. This indicates that these maps should be interpreted as diagnostic internal attention patterns rather than ground-truth localization of size or shape.

Conclusion

This thesis studied the problem of fine-grained hierarchical bird image classification on the CUB-200-2011 dataset. Unlike flat species classification, hierarchical bird recognition requires the model to predict multiple taxonomy levels, including order, family, and species. This setting is challenging because bird species from the same family often share very similar global appearance and differ only in subtle local traits such as bill shape, wing pattern, head pattern, breast color, or tail shape. Therefore, an effective model should capture both hierarchical taxonomic structure and fine-grained visual evidence.

To address this problem, this thesis proposed an attribute-guided extension of H-CAST. The proposed method keeps H-CAST as the image-based hierarchical backbone and adds an attribute-guided learning module at the species-level branch. Instead of predicting all CUB attributes as a single flat vector, the model uses group-specific attribute classifiers based on semantic attribute groups. Each active attribute group is represented by a learnable query, which attends to species-level graph features produced by H-CAST. The resulting attribute-oriented features are used for auxiliary attribute prediction and are fused with the species class token for final species classification. Family and order predictions remain produced by the original H-CAST hierarchical stages.

A key component of the proposed method is the Hard-Sibling Discriminative Attribute Selection strategy. This strategy selects attributes that are useful for distinguishing species from visually similar sibling species within the same family. The goal is to reduce potentially redundant attribute supervision and focus the attribute branch on fine-grained cues that are more relevant to species-level discrimination. In addition, a diversity regularization term is used to encourage attribute-related attention to be less redundant.

Experiments were conducted under the official CUB train/test protocol. Compared with the H-CAST rerun, the proposed attribute-guided H-CAST improved Full-Path Accuracy from 82.86% to 84.36%, species accuracy from 84.55% to 85.83%, family accuracy from 95.22% to 95.62%, order accuracy from 98.74% to 99.03%, and wAP from 86.90% to 88.00%. TICE was also reduced from 4.35% to 3.62%. These results show that the proposed attribute-guided learning strategy provides useful additional supervision for hierarchical fine-grained bird classification.

The ablation studies further showed that the best FPA, species accuracy, and wAP

among the fusion-design settings are obtained when attribute supervision, species-level fusion, and diversity regularization are used together. Attribute supervision alone, without species-level fusion, was not sufficient to improve species classification. Conversely, fusion without attribute supervision improved classification to some extent, but the attribute branch was not semantically aligned with CUB attributes. This confirms that both semantic attribute supervision and explicit fusion are important for the proposed method. The diversity loss improved species accuracy and wAP at a moderate weight, although it introduced a trade-off with taxonomy consistency.

The analysis of attribute selection also showed that the proposed HSDAS-199 setting achieved the best species accuracy and wAP among the evaluated attribute-selection variants. However, random attribute subsets remained competitive in some metrics, especially TICE and family accuracy. Therefore, the proposed attribute selection strategy should not be interpreted as dominating all alternatives in every metric. Instead, the results support HSDAS as a targeted and interpretable strategy that provides useful bias toward species-level discrimination.

The qualitative and diagnostic analysis provided additional insight into the model behavior. The proposed model corrected 236 species-level errors made by the H-CAST rerun, while introducing 161 new species-level errors where H-CAST was originally correct. This gives a net gain of 75 correctly classified test images, which is consistent with the improvement in species accuracy. Among the corrected cases, 194 were intra-family confusions, suggesting that many improvements occur in fine-grained cases between visually related species. At the same time, attention visualizations showed that the model should not be interpreted as a precise attribute localization model. Global attribute groups such as size and shape often received high pooling weights, and their spatial maps could be diffuse or include background regions. Thus, the attention maps were treated only as diagnostic evidence rather than ground-truth localization.

Despite the improvements, the proposed method has several limitations. First, it depends on the availability and quality of class-level attribute annotations. Second, the attribute-guided fusion is mainly applied to the species-level branch, while family and order predictions are not directly fused with attribute-oriented features. Third, the method does not use pixel-level attribute localization supervision, so the attention maps may reflect global morphology, silhouette, context, or background cues in addition to object-part evidence. Finally, the performance depends on the selected attribute subset and loss

weights.

Future work can extend this study in several directions. One direction is to incorporate object masks, part annotations, or localization constraints to encourage attribute queries to focus more consistently on bird regions. Another direction is to improve taxonomy consistency by applying attribute-guided fusion not only at the species level but also at higher hierarchy levels. Further work can also investigate more robust attribute selection strategies, evaluate the method on additional fine-grained hierarchical datasets, and study whether attribute-guided representations improve interpretability under stronger localization supervision.

In summary, this thesis demonstrates that incorporating semantic attribute guidance into an H-CAST-based hierarchical classifier can improve fine-grained bird species recognition. The proposed method is most effective when attribute supervision is combined with species-level fusion and moderate diversity regularization. While the method does not provide reliable attribute localization, it improves hierarchical classification performance and offers a useful direction for combining visual grouping, taxonomy structure, and semantic attribute supervision in fine-grained recognition.

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